# FALCUTÉ DES SCIENCES ET TECHNOLOGIES (FST)

## TROISIÈME ANNÉE

Rapport du travail de Laboratoire N$^{0}$ 1

Cours : Reseaux

Étudiant : BELDOUETTE Darvens Mardy Sharkyl

Professeur : Ismaël SAINT AMOUR

**LE 25 OCTOBRE 2026**

# 1. DESCRIPTION DES RESULTATS DE LA TACHE

### 1.1 Objectifs du travail dirigé

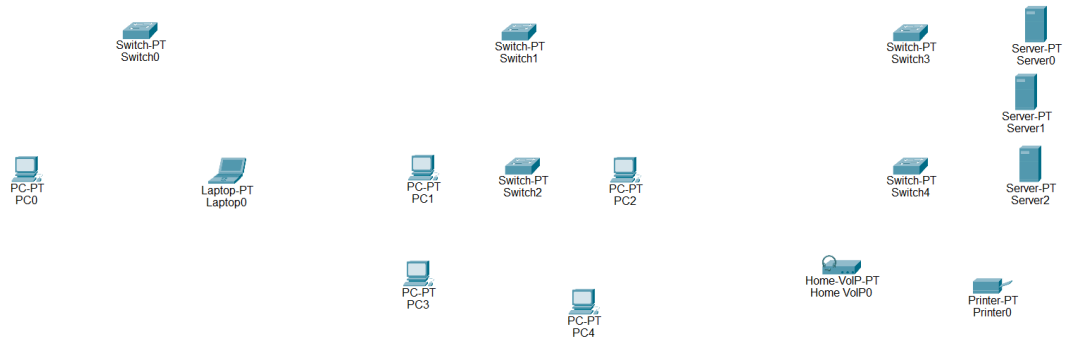
Ce quatrième travail dirigé porte sur la mise en pratique de la configuration des équipements d'interconnexion sous Cisco Packet Tracer, en lien avec les fondements physiques de la transmission étudiés dans le Module 4 (Signaux numériques, support de transmission guidé, débit binaire et propagation des trames). Les objectifs spécifiques consistent à :

- Découvrir et appliquer la configuration de base d'un commutateur (Switch Cisco 2960-24TT) et d'un routeur (Router Cisco 2811) en ligne de commande (CLI).
- Attribuer des adresses IPv4 et IPv6 aux interfaces d'administration des switchs (VLAN 1), aux interfaces du routeur et aux hôtes finaux (PC, Laptop, serveurs, imprimante et téléphone VoIP).
- Vérifier la connectivité de bout en bout entre les différents segments du réseau (sites Jacmel 1 et Jacmel 2) à l'aide de la commande `ping`.
- Utiliser le Mode Simulation pour observer l'encapsulation, le cheminement des paquets ICMP et les échanges sur les supports de transmission.

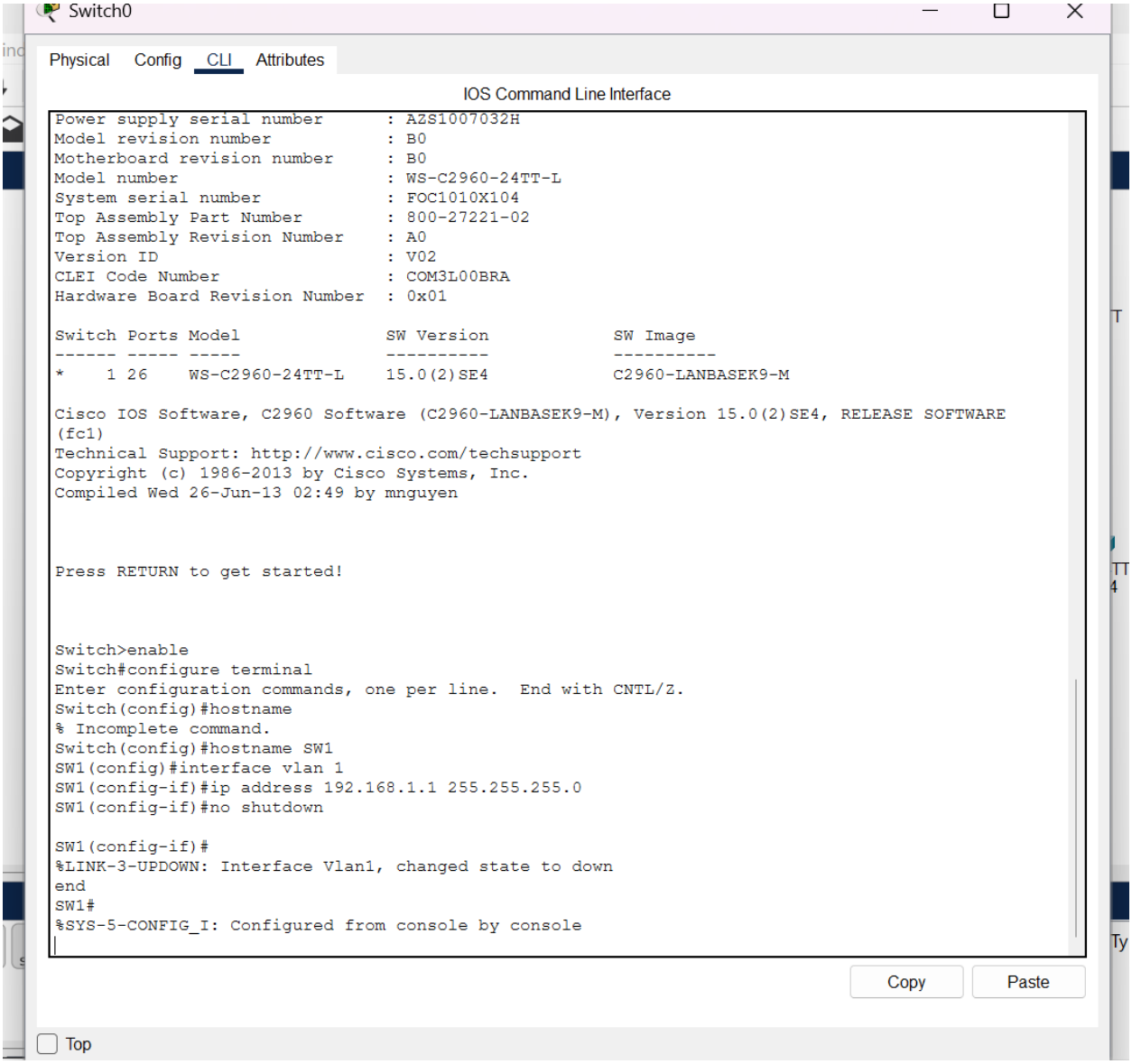
### 1.3 Démarche suivie

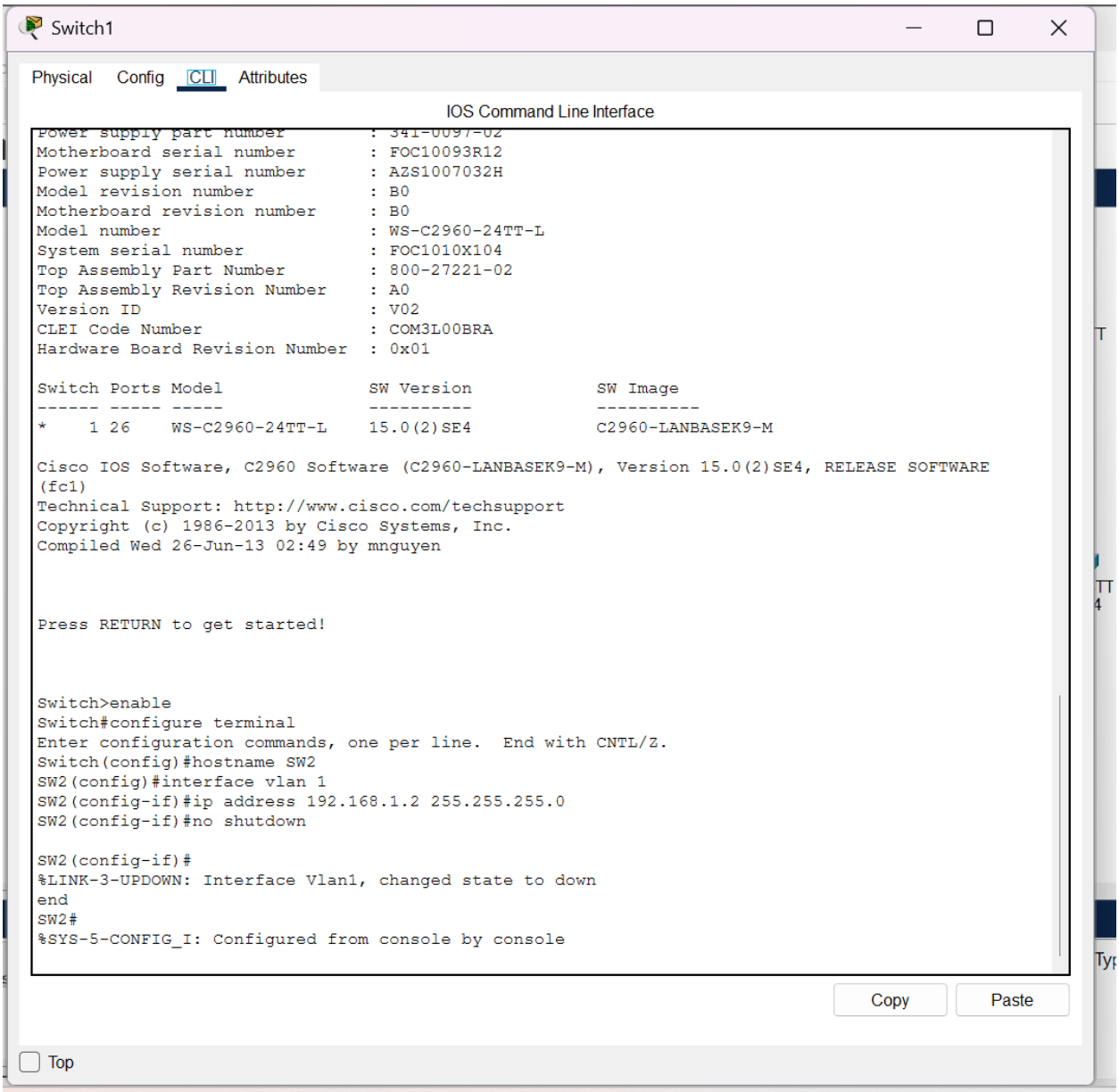
1. **Construction des topologies :** Reproduction des architectures sous Packet Tracer en reliant les périphériques terminaux aux commutateurs par des câbles droits, et les commutateurs entre eux par des câbles croisés, puis intégration du routeur Cisco 2811 entre les deux zones réseau (Jacmel 1 et Jacmel 2).
2. **Configuration des commutateurs (SW1 / SW2) :** Accès au terminal CLI, passage en mode privilégié puis global, attribution du nom d'hôte, configuration d'une adresse IPv4 et IPv6 sur l'interface logique `VLAN 1`, activation par `no shutdown` et définition de la passerelle par défaut (`ip default-gateway`).
3. **Configuration du routeur (R1 / Router0) :** Attribution d'un nom d'hôte et adressage des interfaces physiques `FastEthernet0/0` et `FastEthernet0/1` servant de passerelles aux sous-réseaux interconnectés.
4. **Adressage des hôtes et validation :** Attribution manuelle des adresses IPv4/IPv6 et passerelles sur chaque poste, exécution de tests `ping` en mode Temps Réel et suivi pas à pas des unités de données de protocole (PDU) en Mode Simulation.

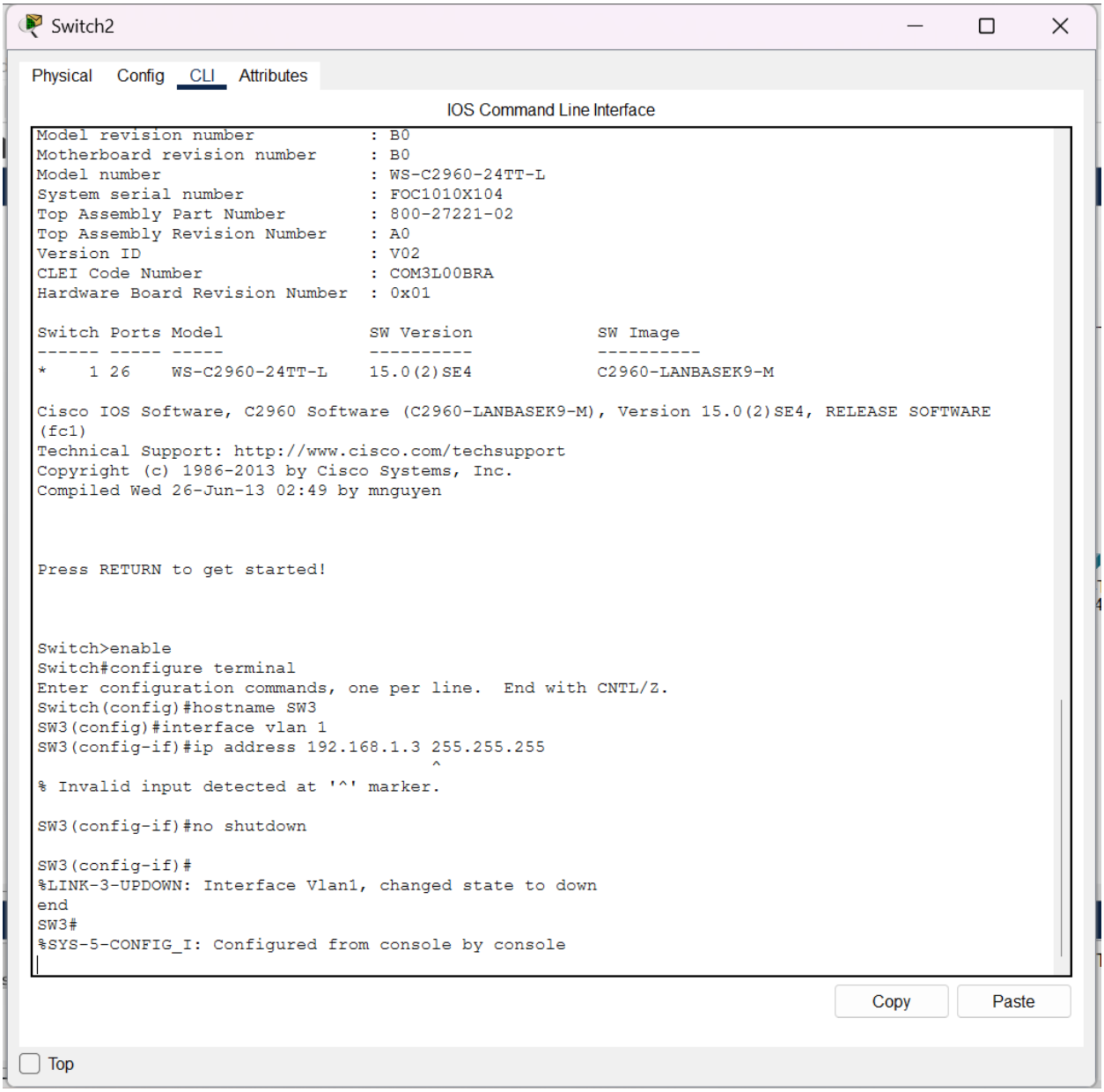
# EXERCICE 1

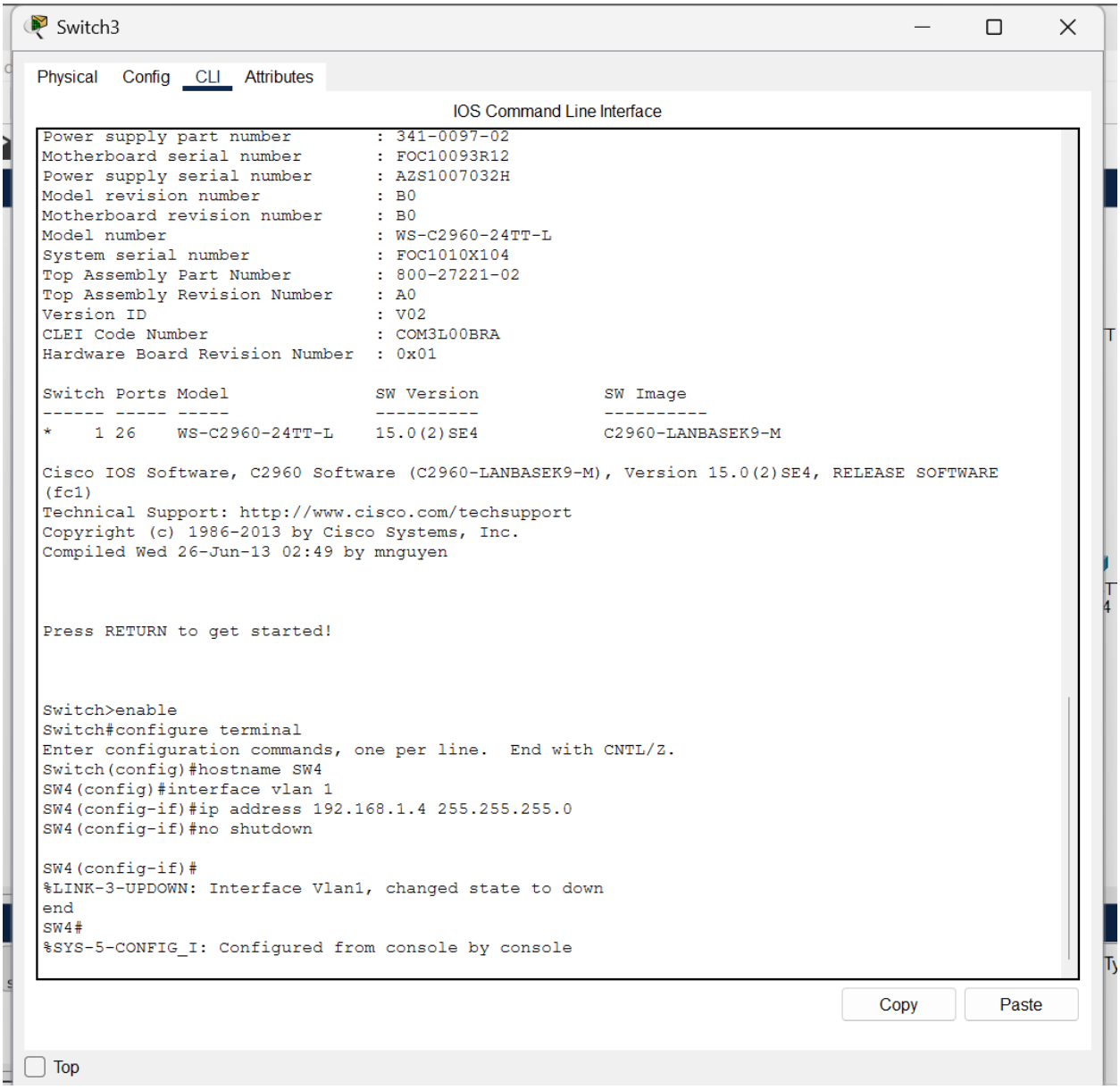


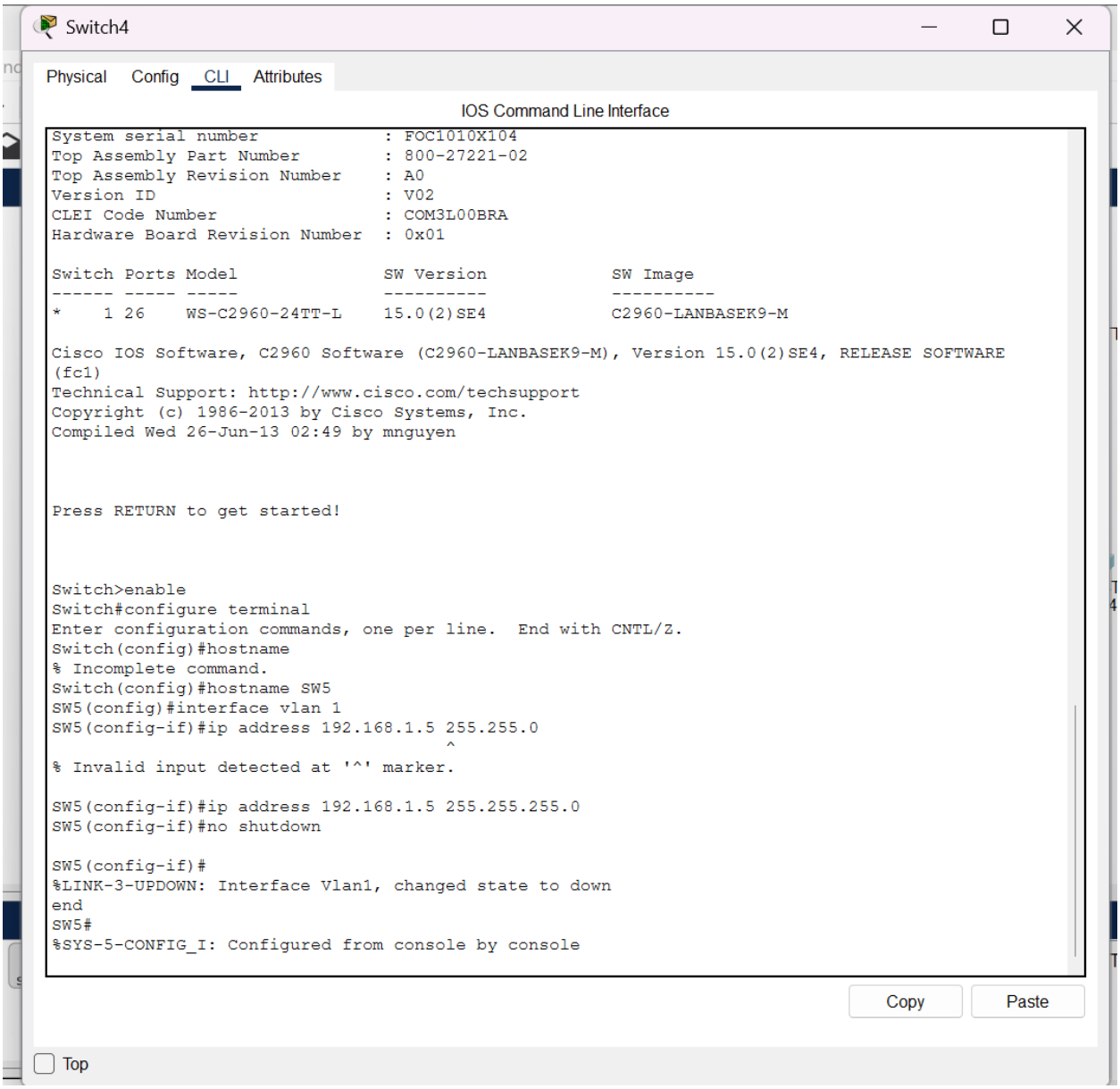

## Configuration des switchs IPv4



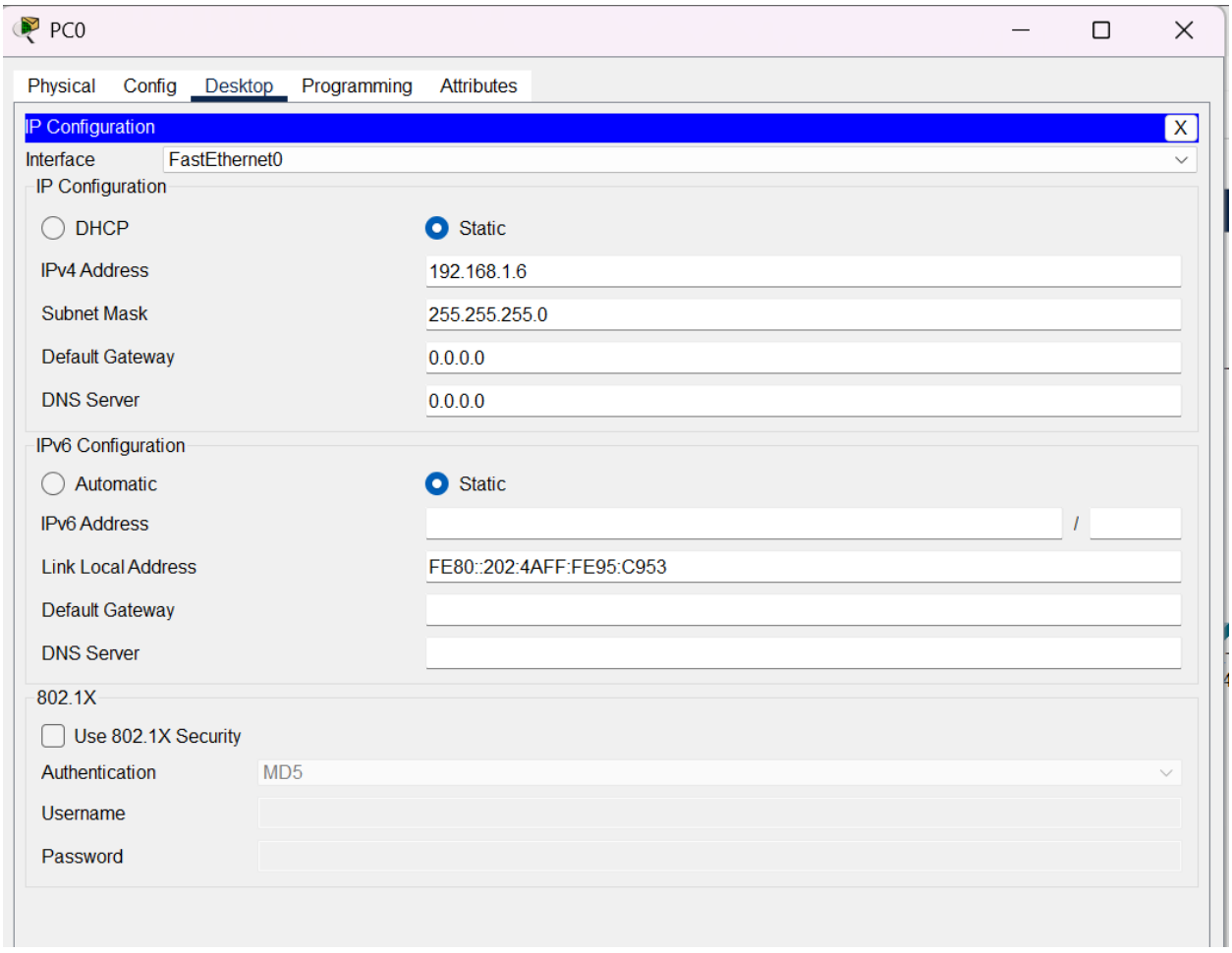

Switch1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Power supply part number : 341-0097-02
Motherboard serial number : FOC10093R12
Power supply serial number : AZS1007032H
Model revision number : B0
Motherboard revision number : B0
Model number : WS-C2960-24TT-L
System serial number : FOC1010X104
Top Assembly Part Number : 800-27221-02
Top Assembly Revision Number : A0
Version ID : V02
CLEI Code Number : COM3L00BRA
Hardware Board Revision Number : 0x01
Switch Ports Model SW Version SW Image
------ ----- ----- ---------- ----------
* 1 26 WS-C2960-24TT-L 15.0(2)SE4 C2960-LANBASEK9-M
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE
(fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen
Press RETURN to get started!
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW2
SW2(config)#interface vlan 1
SW2(config-if)#ip address 192.168.1.2 255.255.255.0
SW2(config-if)#no shutdown
SW2(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
end
SW2#
%SYS-5-CONFIG_I: Configured from console by console
Copy
Paste
Top

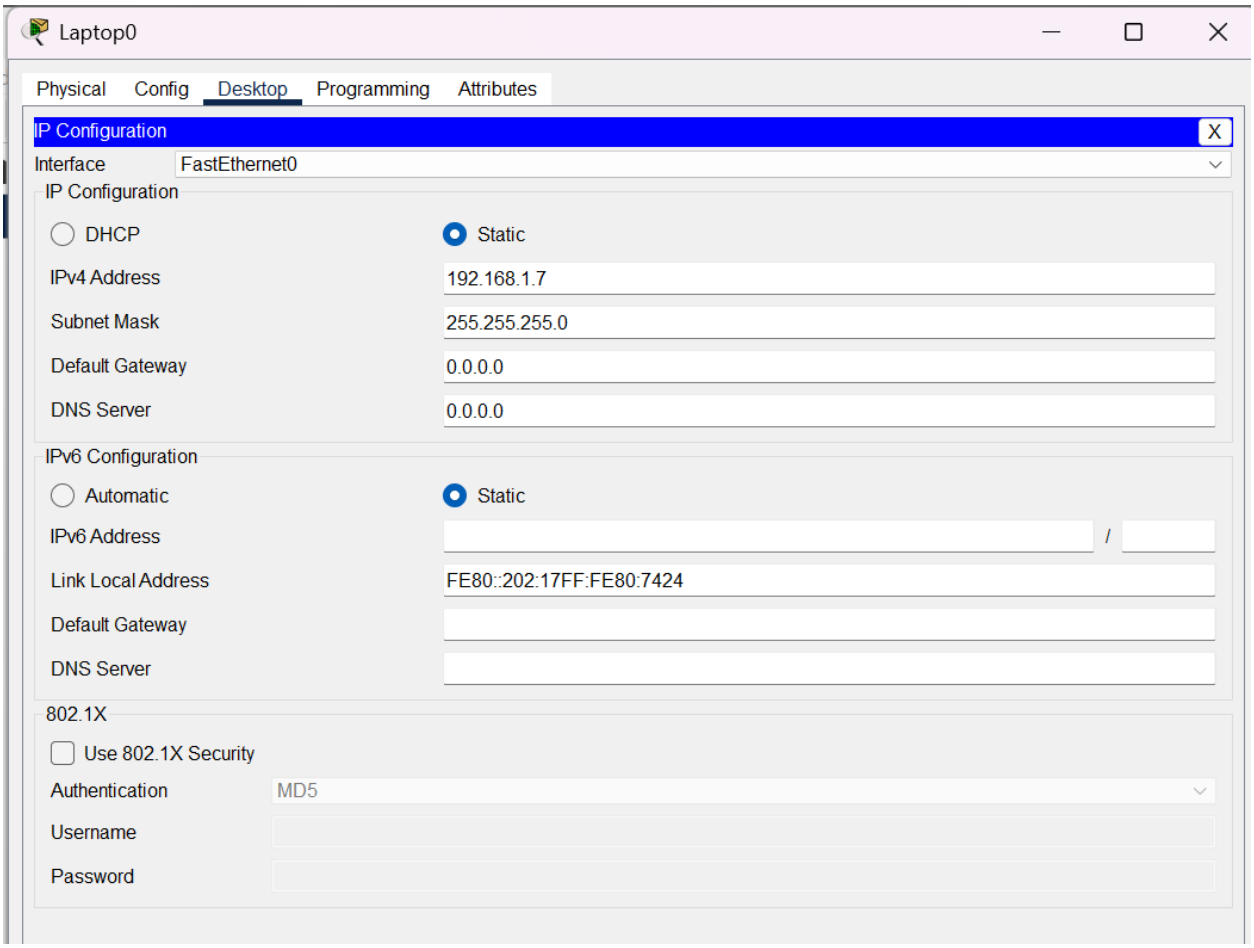

Switch2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Model revision number : B0
Motherboard revision number : B0
Model number : WS-C2960-24TT-L
System serial number : FOC1010X104
Top Assembly Part Number : 800-27221-02
Top Assembly Revision Number : A0
Version ID : V02
CLEI Code Number : COM3L00BRA
Hardware Board Revision Number : 0x01
Switch Ports Model SW Version SW Image
* 1 26 WS-C2960-24TT-L 15.0(2)SE4 C2960-LANBASEK9-M
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen
Press RETURN to get started!
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW3
SW3(config)#interface vlan 1
SW3(config-if)#ip address 192.168.1.3 255.255.255
^
% Invalid input detected at '^' marker.
SW3(config-if)#no shutdown
SW3(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
end
SW3#
%SYS-5-CONFIG_I: Configured from console by console
Copy
Paste
Top

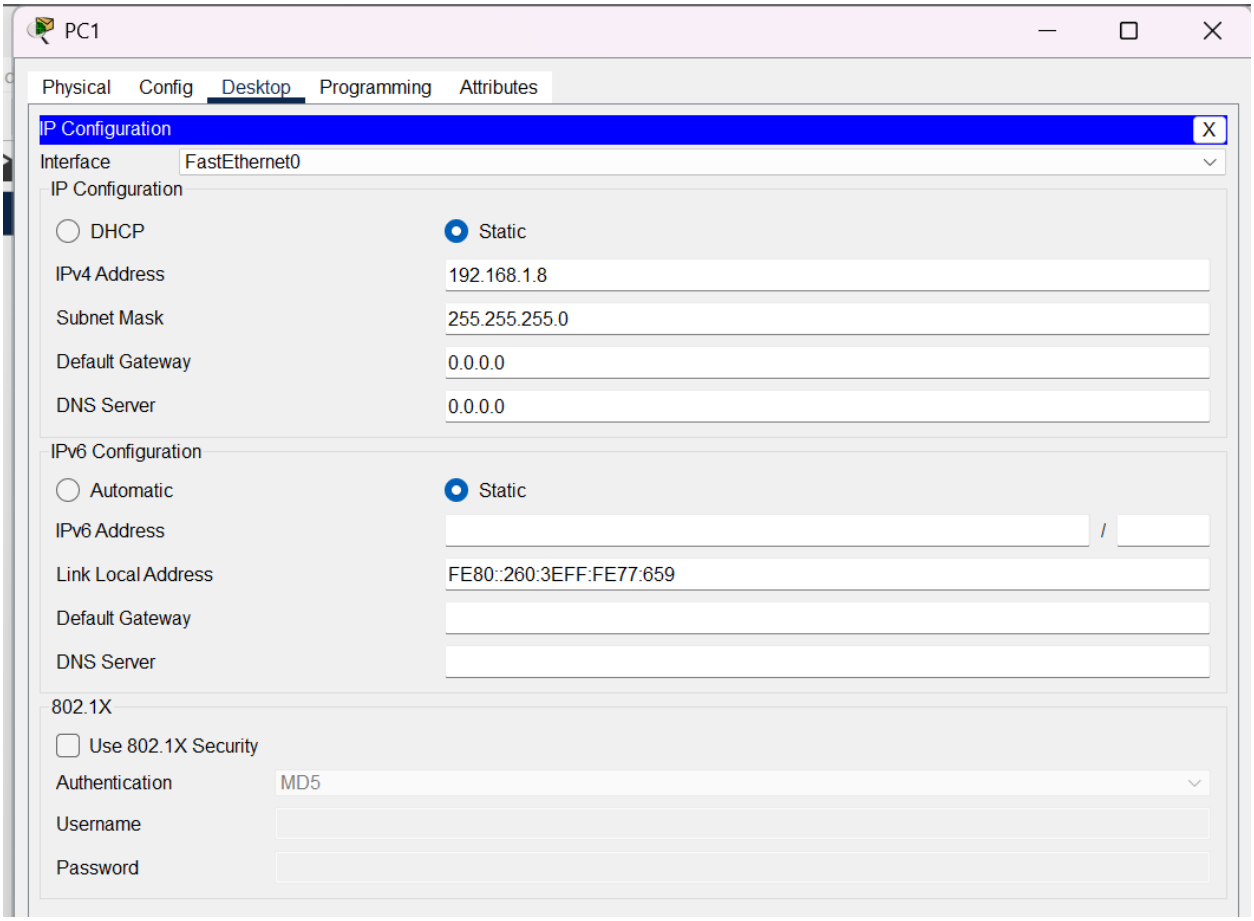

Switch3
Physical
Config
CLI
Attributes
IOS Command Line Interface
Power supply part number : 341-0097-02
Motherboard serial number : FOC10093R12
Power supply serial number : AZS1007032H
Model revision number : B0
Motherboard revision number : B0
Model number : WS-C2960-24TT-L
System serial number : FOC1010X104
Top Assembly Part Number : 800-27221-02
Top Assembly Revision Number : A0
Version ID : V02
CLEI Code Number : COM3L00BRA
Hardware Board Revision Number : 0x01
Switch Ports Model SW Version SW Image
* 1 26 WS-C2960-24TT-L 15.0(2)SE4 C2960-LANBASEK9-M
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen
Press RETURN to get started!
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW4
SW4(config)#interface vlan 1
SW4(config-if)#ip address 192.168.1.4 255.255.255.0
SW4(config-if)#no shutdown
SW4(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
end
SW4#
%SYS-5-CONFIG_I: Configured from console by console
Copy
Paste
Top

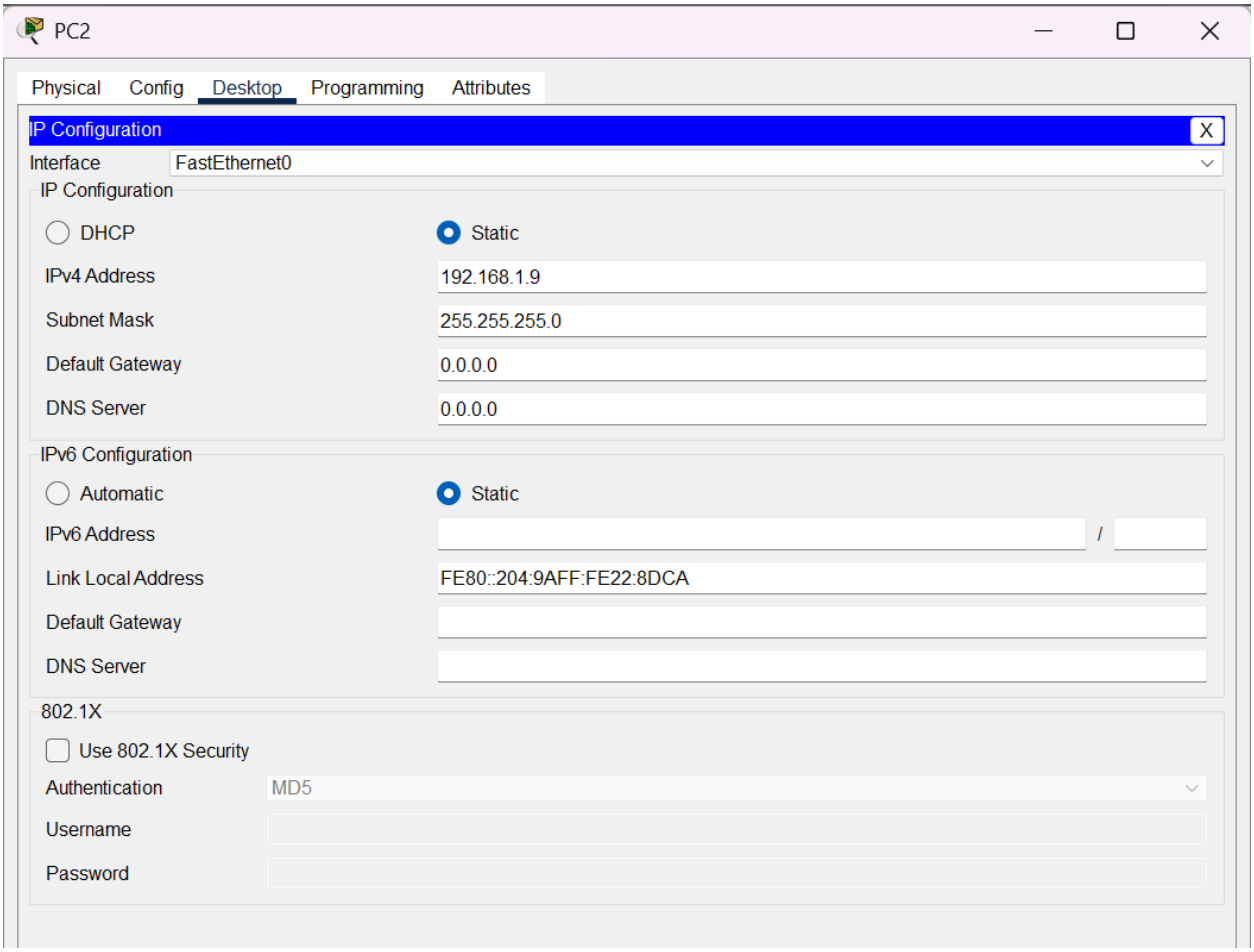

Switch4
Physical
Config
CLI
Attributes
IOS Command Line Interface
System serial number : FOC1010X104
Top Assembly Part Number : 800-27221-02
Top Assembly Revision Number : A0
Version ID : V02
CLEI Code Number : COM3L00BRA
Hardware Board Revision Number : 0x01
Switch Ports Model SW Version SW Image
------ ----- ----- ---------- ----------
* 1 26 WS-C2960-24TT-L 15.0(2)SE4 C2960-LANBASEK9-M
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE
(fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen
Press RETURN to get started!
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname
% Incomplete command.
Switch(config)#hostname SW5
SW5(config)#interface vlan 1
SW5(config-if)#ip address 192.168.1.5 255.255.0
^
% Invalid input detected at '^' marker.
SW5(config-if)#ip address 192.168.1.5 255.255.255.0
SW5(config-if)#no shutdown
SW5(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
end
SW5#
%SYS-5-CONFIG_I: Configured from console by console
Copy
Paste
Top

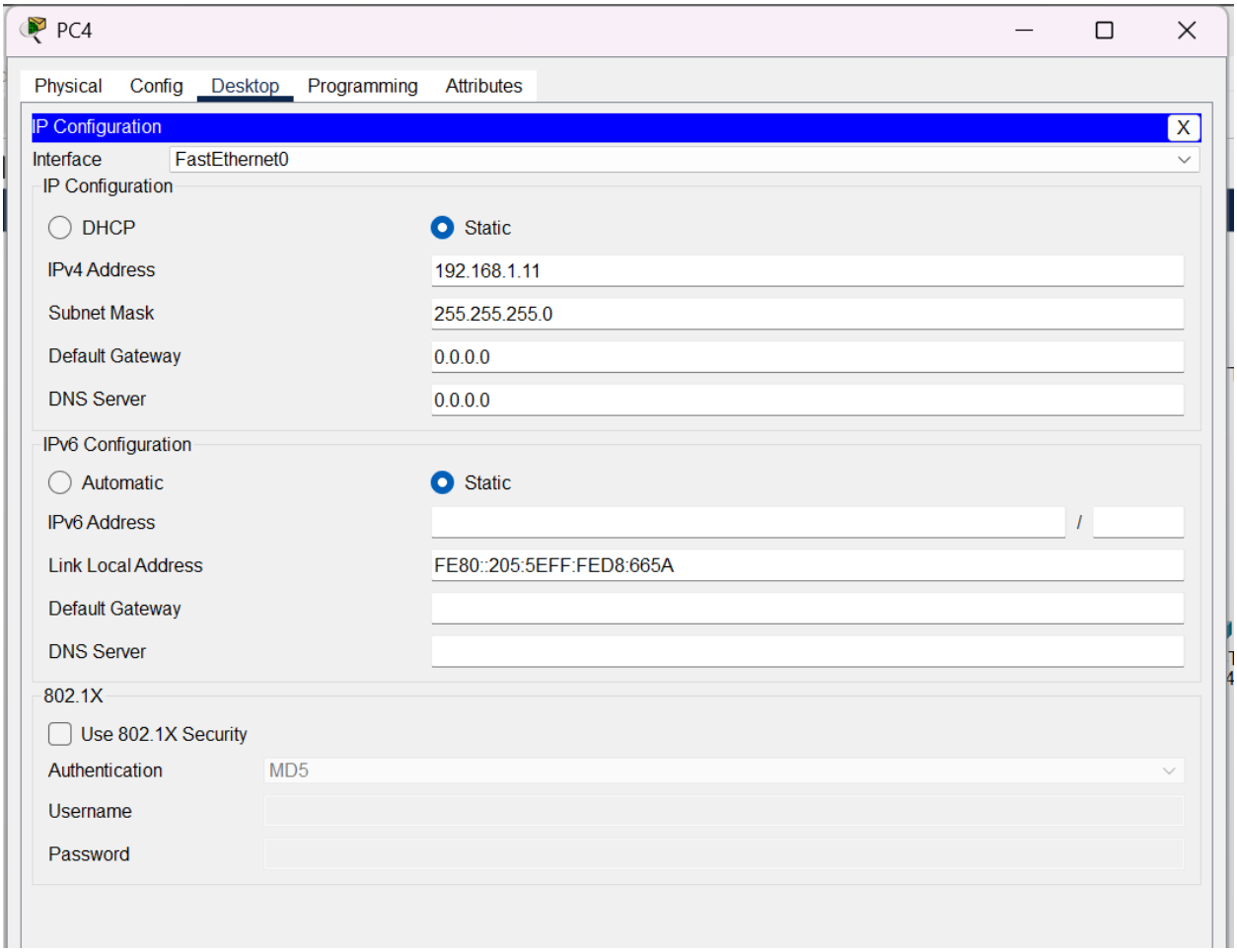
## Attribuons des adresses IPv4 aux périphériques

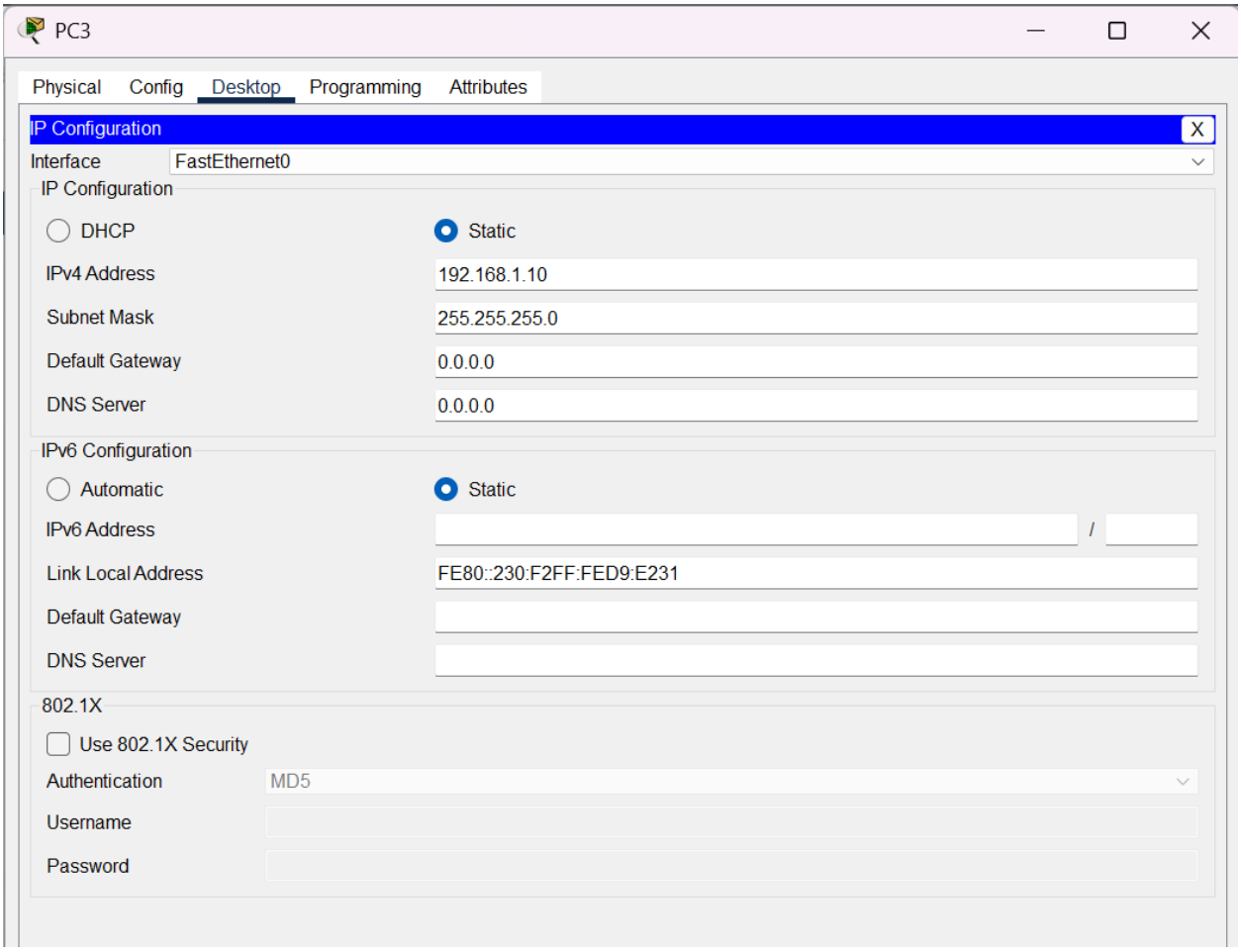


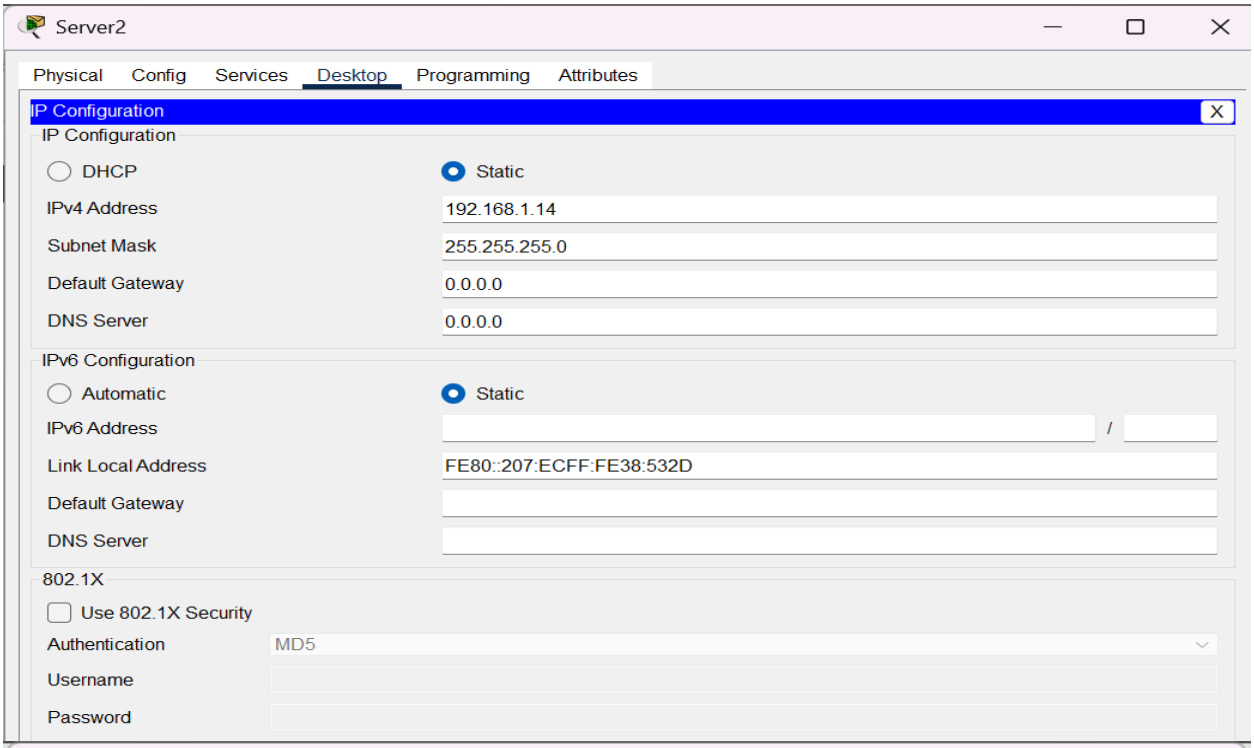

Laptop0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.7
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::202:17FF:FE80:7424
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

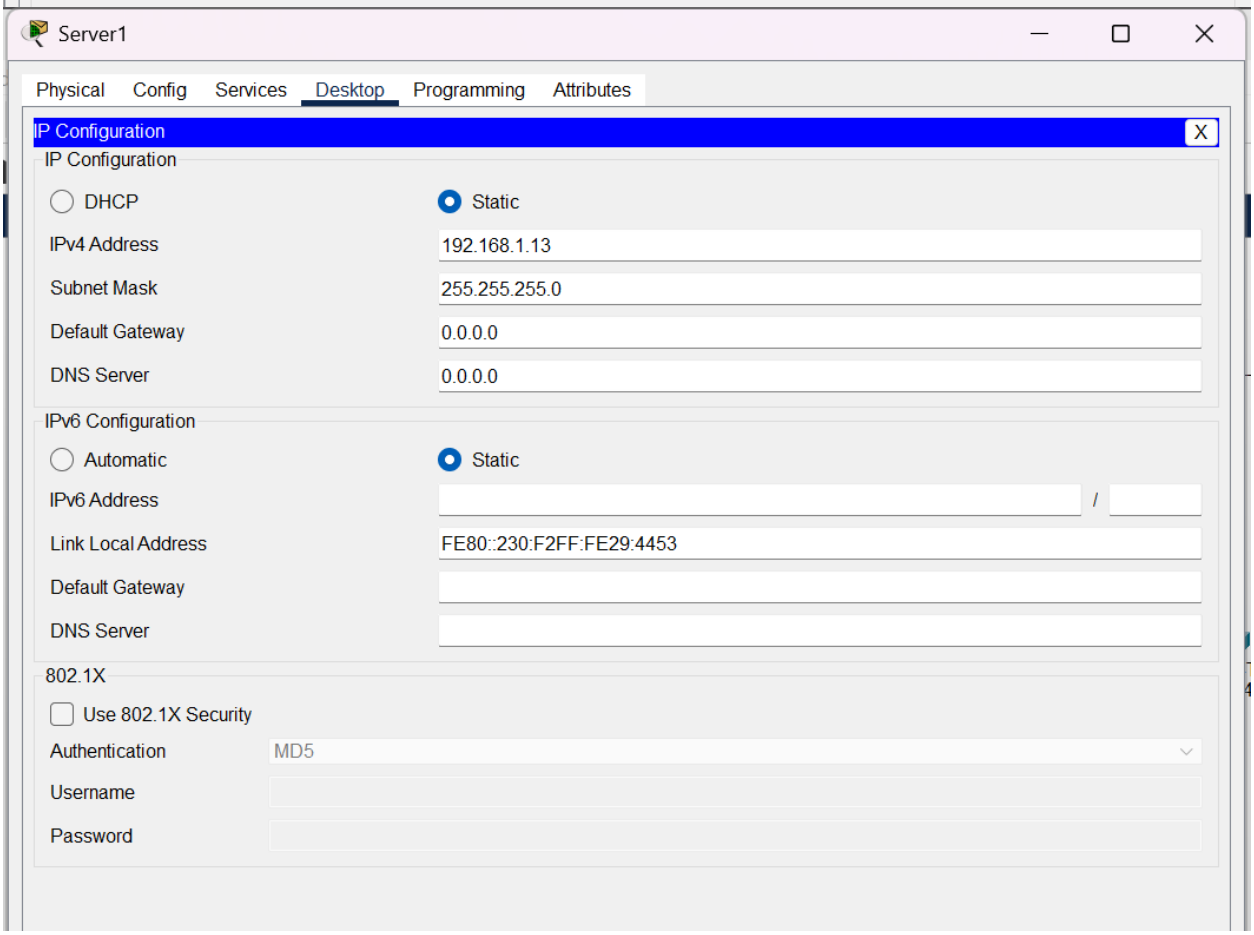

PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.8
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::260:3EFF:FE77:659
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

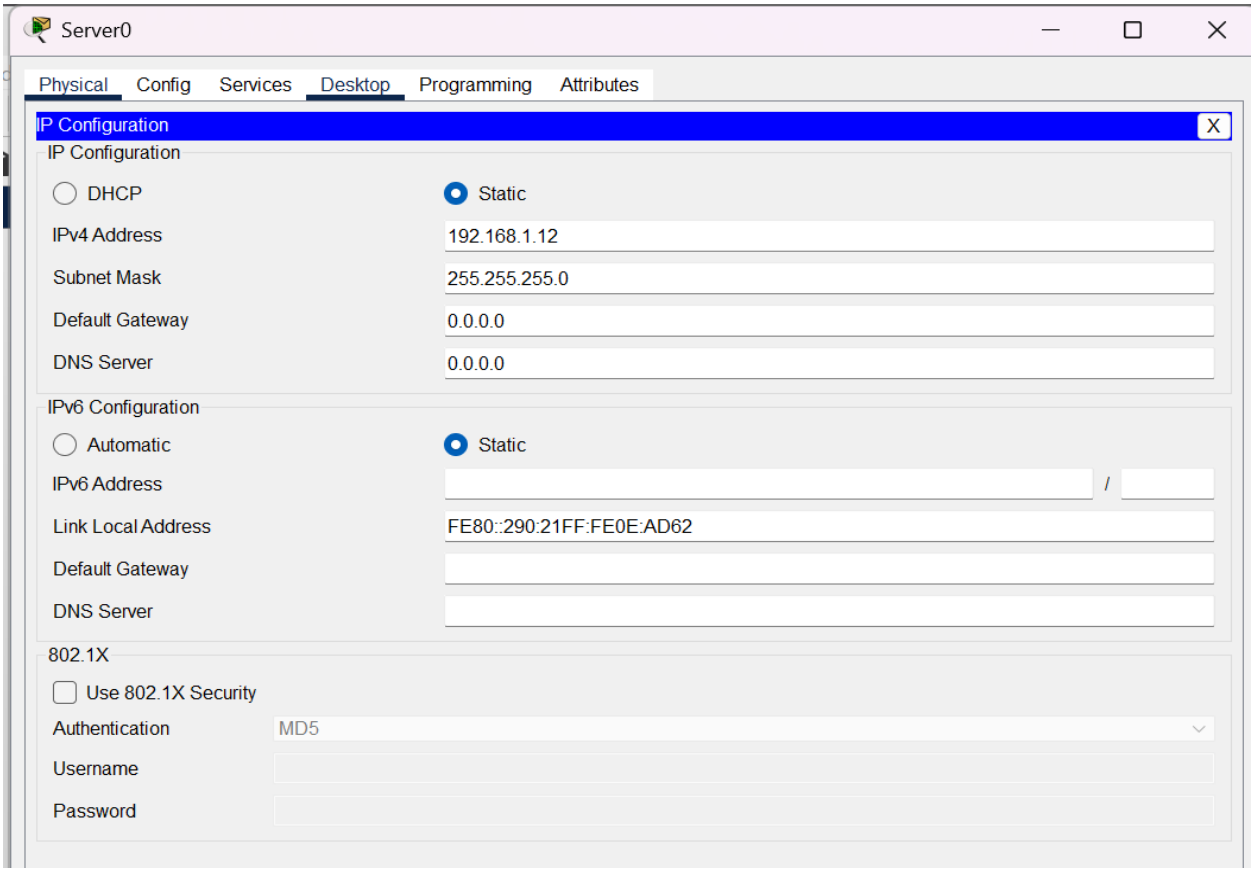

PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.9
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::204:9AFF:FE22:8DCA
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

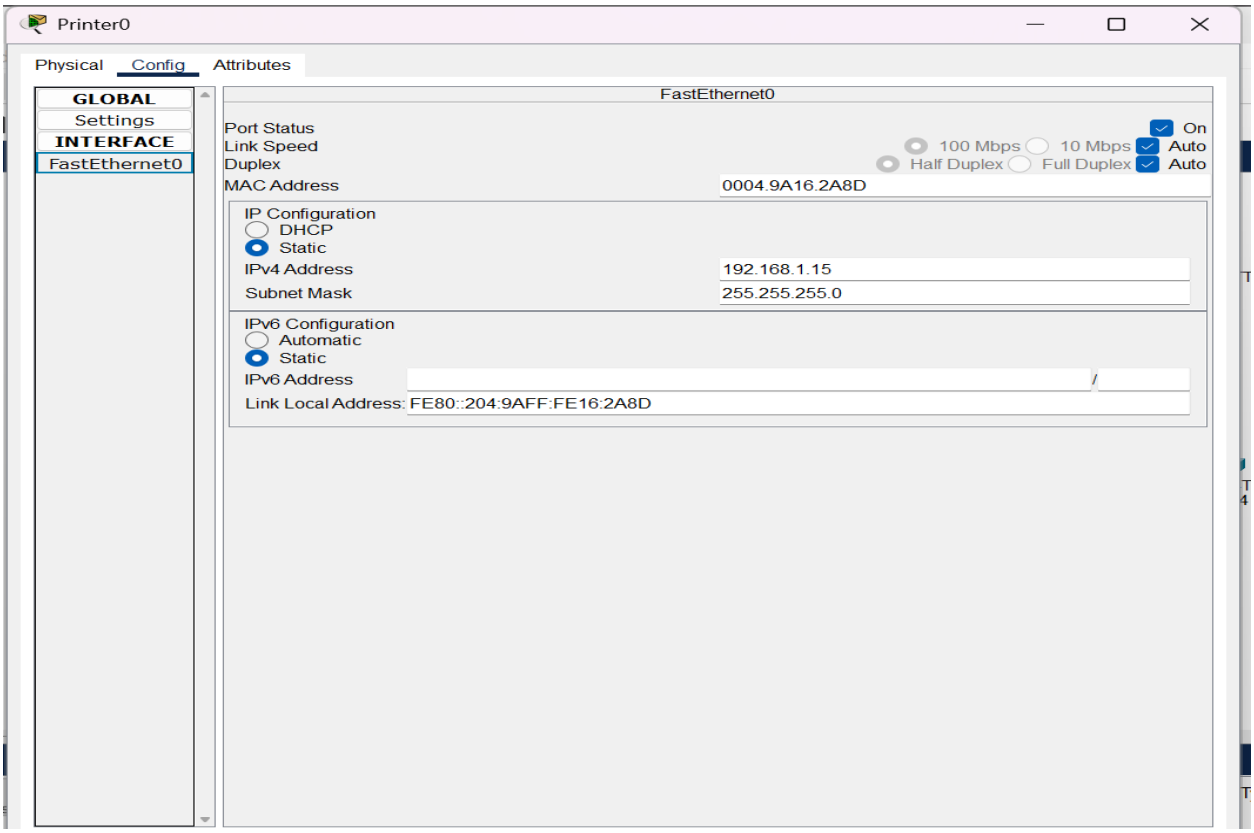

PC4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.11
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::205:5EFF:FED8:665A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.10
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::230:F2FF:FED9:E231
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

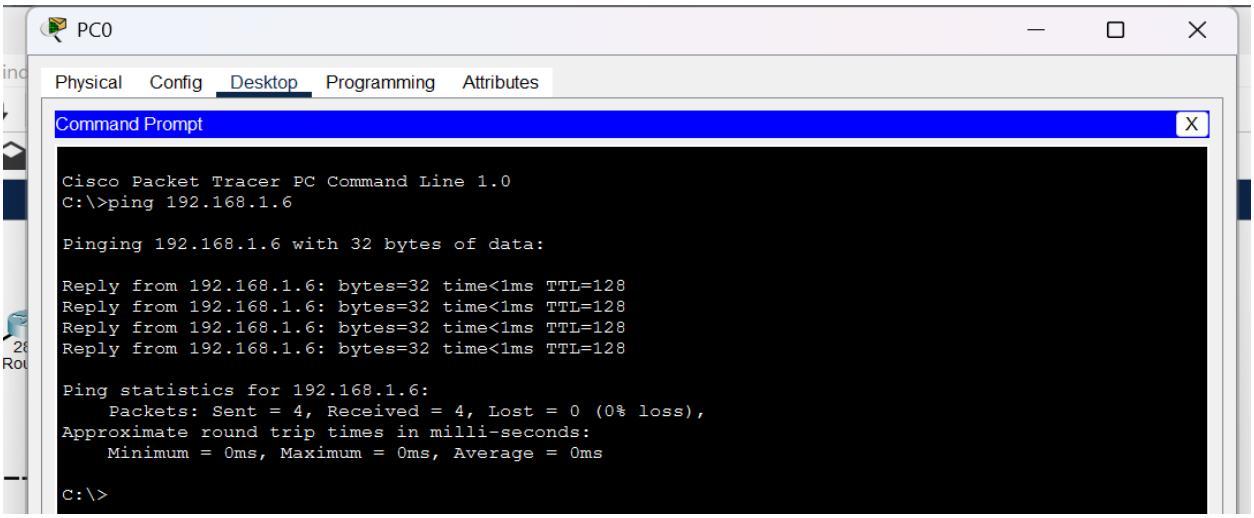

Server2
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.14
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::207:ECFF:FE38:532D
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

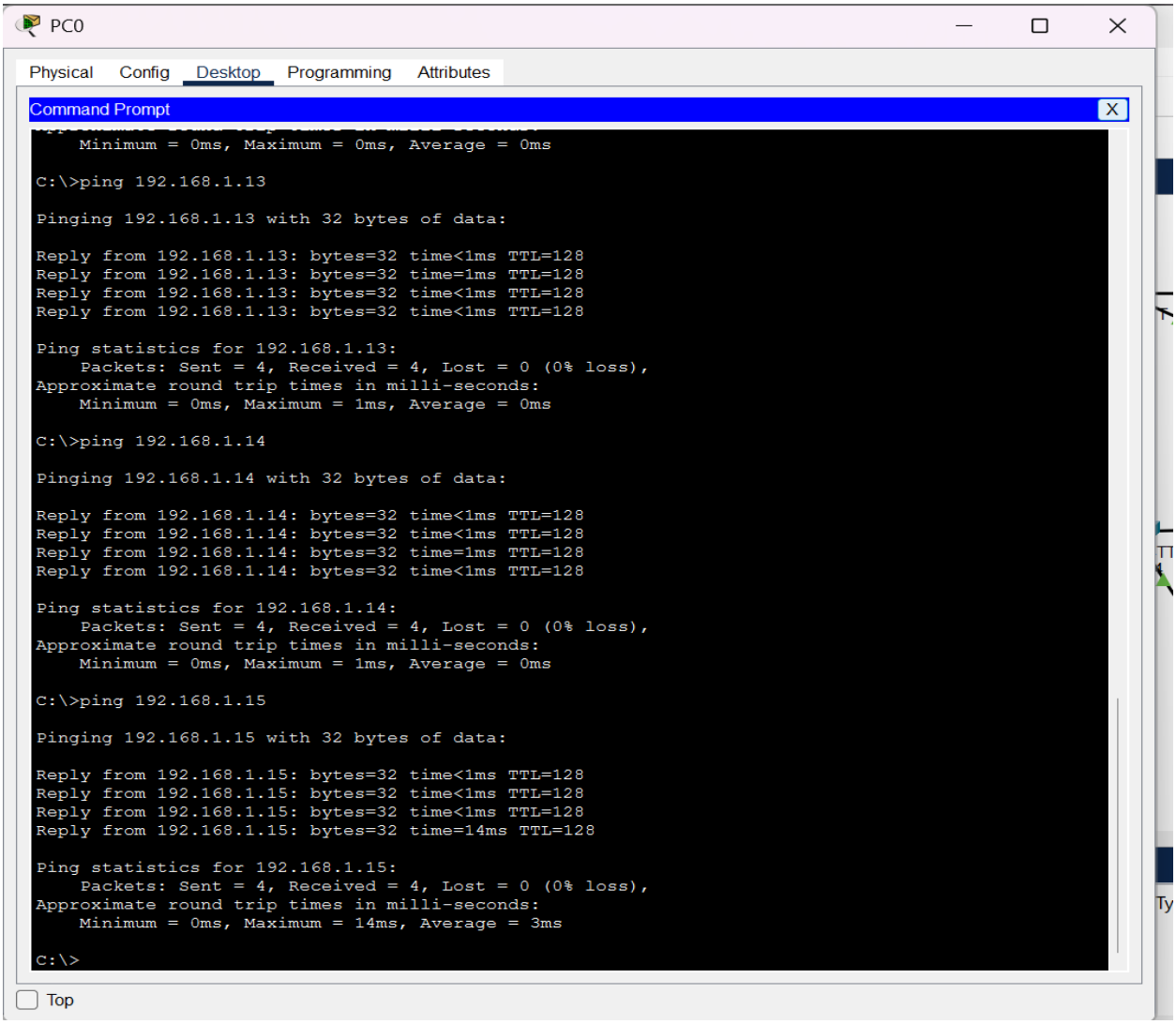

Server1
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.13
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::230:F2FF:FE29:4453
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


Server0
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.12
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::290:21FF:FE0E:AD62
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

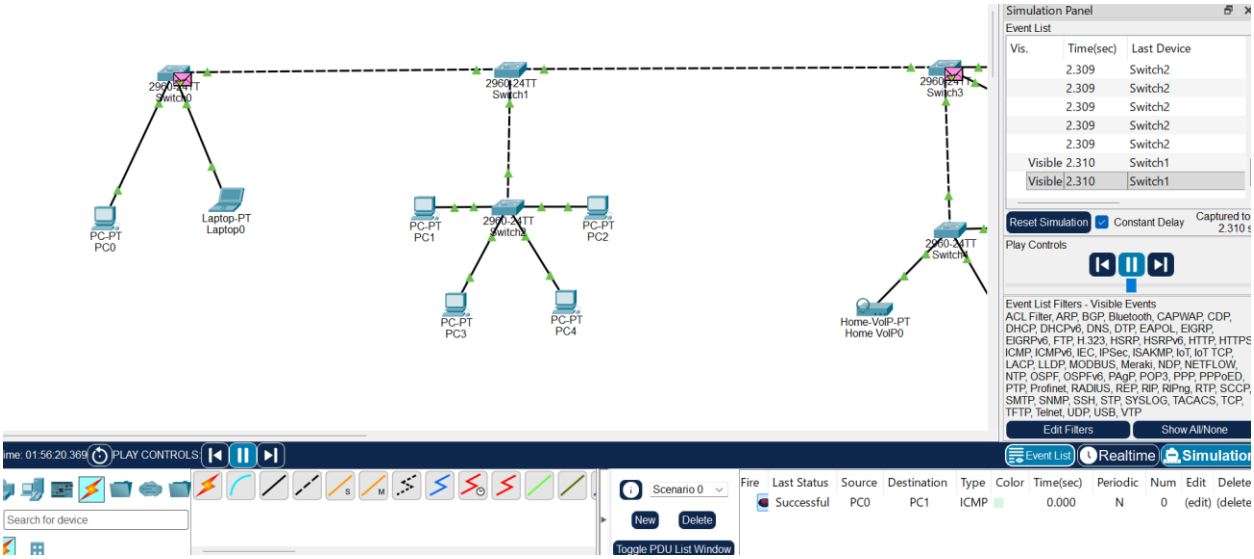

Printer0
Physical
Config
Attributes
GLOBAL
Settings
INTERFACE
FastEthernet0
FastEthernet0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0004.9A16.2A8D
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.15
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::204:9AFF:FE16:2A8D

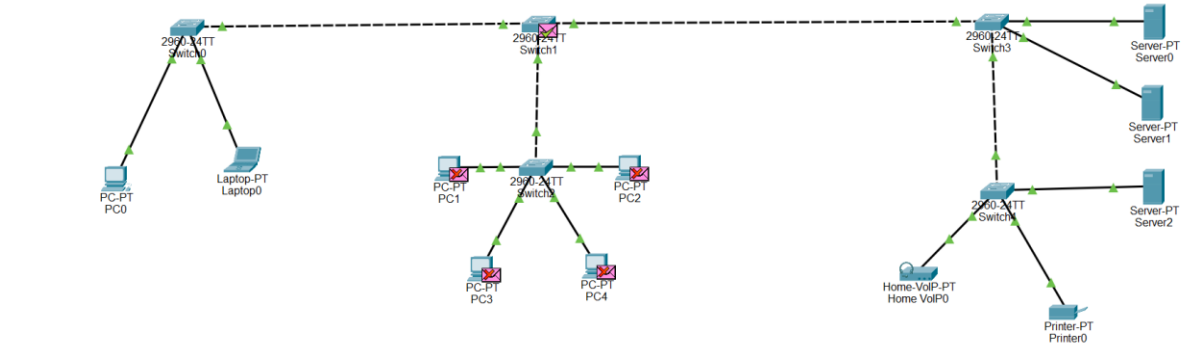
## Tester la connectivité entre les hôtes.

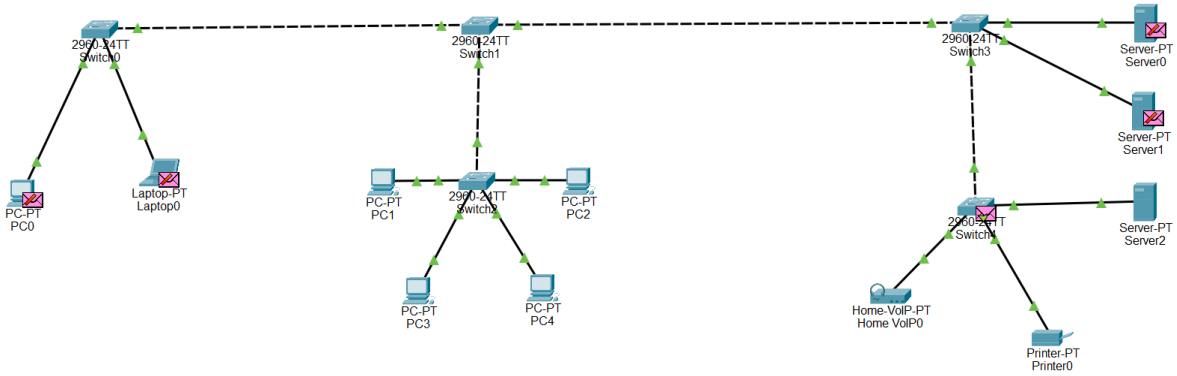


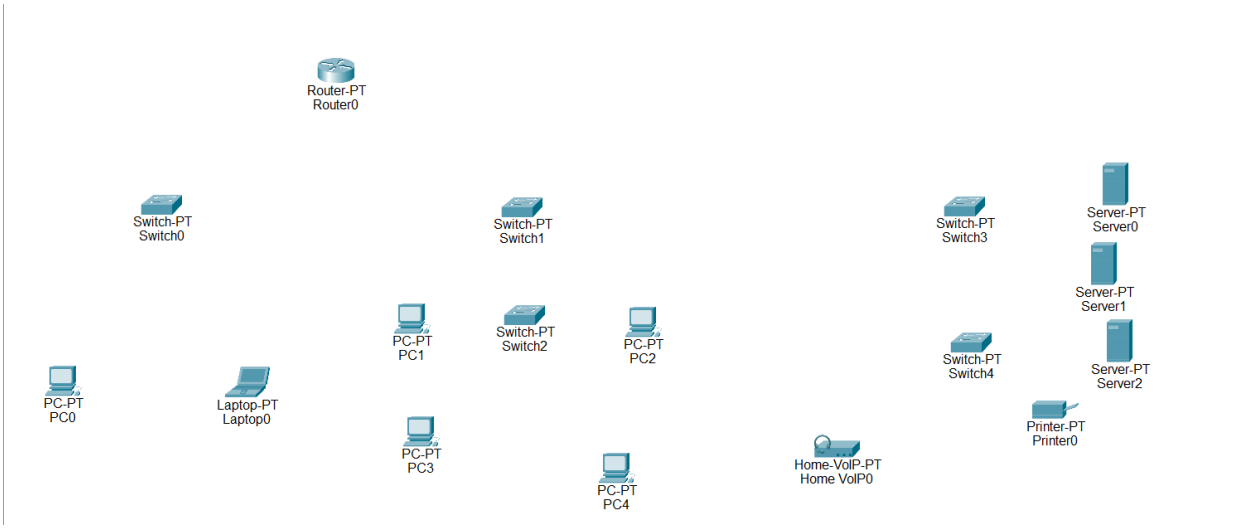
# Utiliser le Mode Simulation pour analyser le trafic réseau.



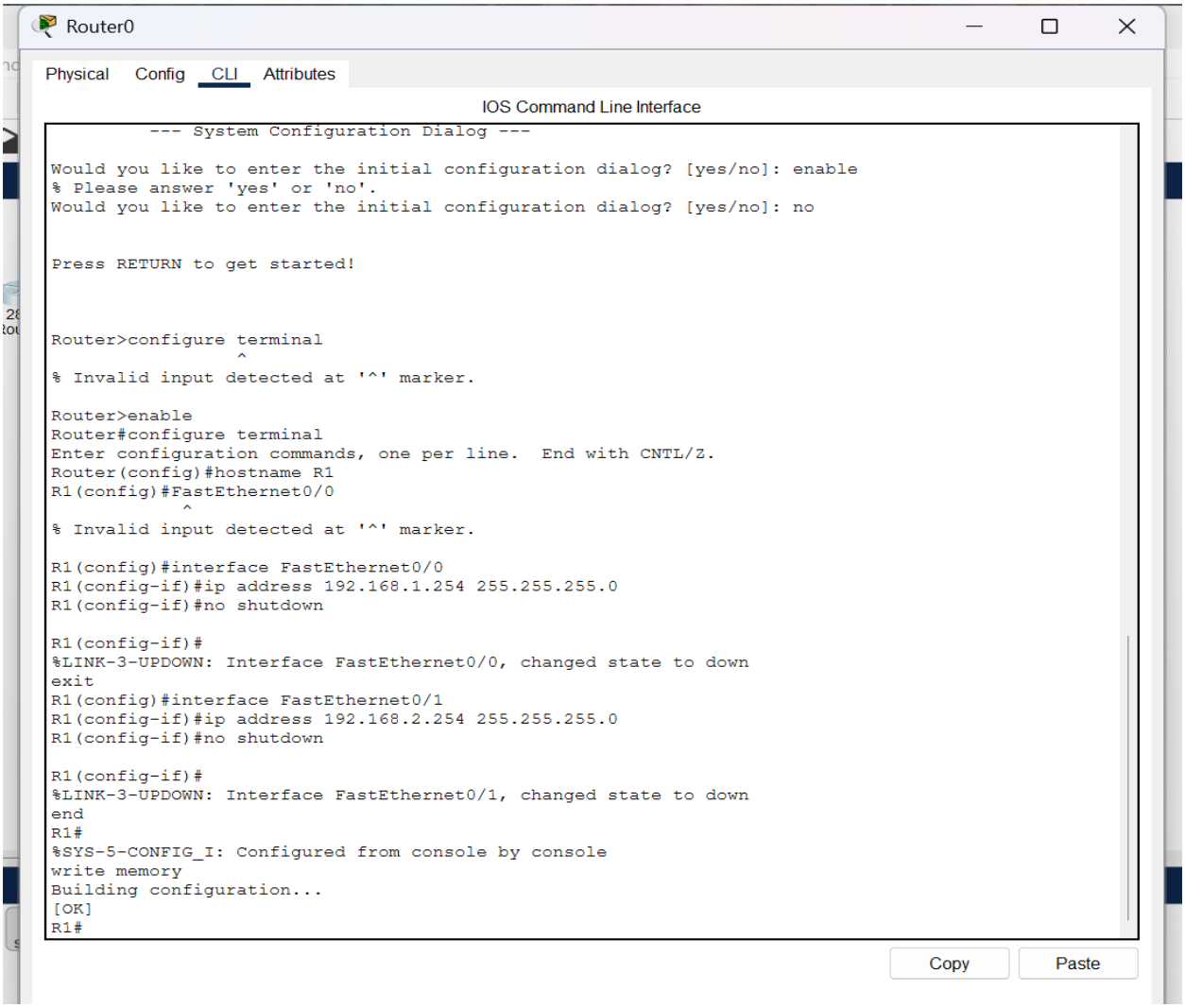



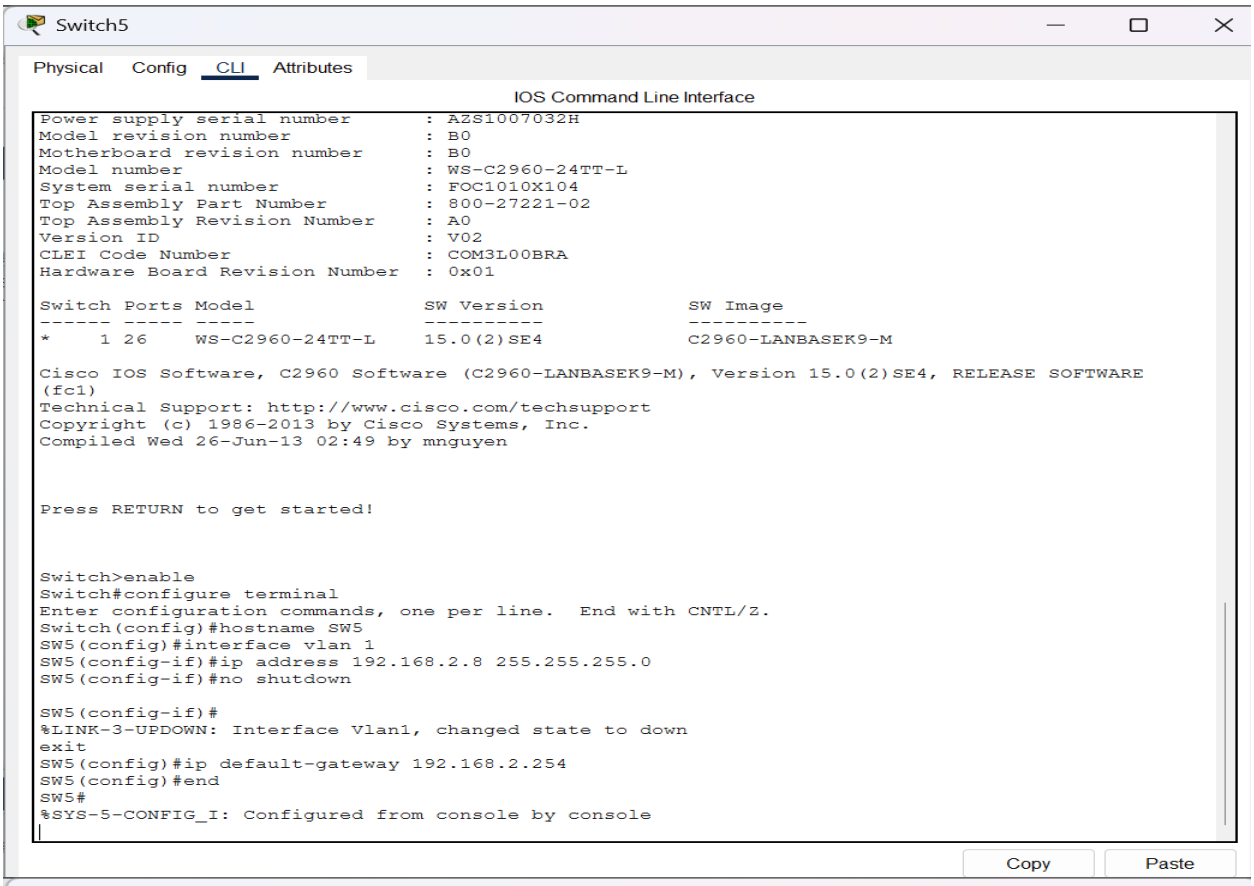



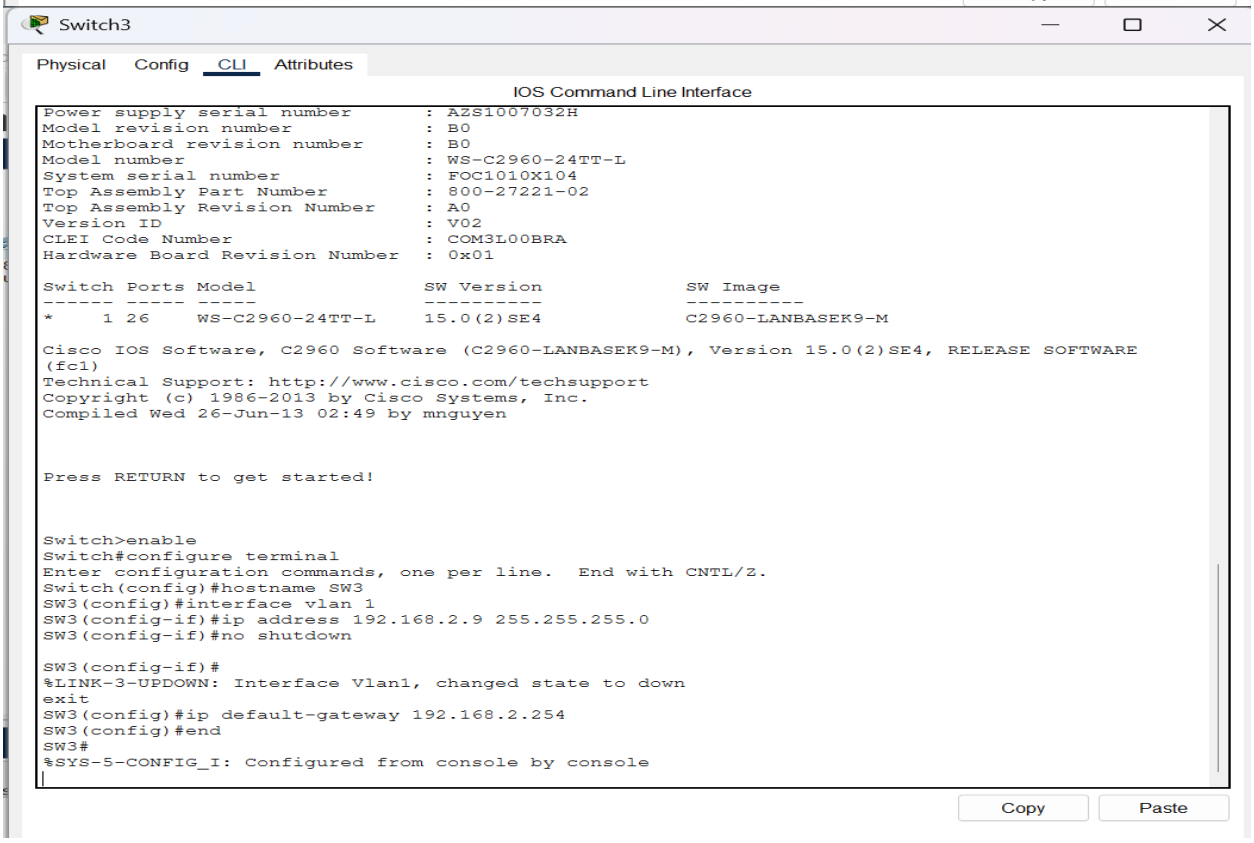
# EXERCICE 2




## Configuration IPv4 du routeur et des switchs




Switch5
Physical
Config
CLI
Attributes
IOS Command Line Interface
Power supply serial number : AZS1007032H
Model revision number : B0
Motherboard revision number : B0
Model number : WS-C2960-24TT-L
System serial number : FOC1010X104
Top Assembly Part Number : 800-27221-02
Top Assembly Revision Number : A0
Version ID : V02
CLEI Code Number : COM3L00BRA
Hardware Board Revision Number : 0x01
Switch Ports Model SW Version SW Image
* 1 26 WS-C2960-24TT-L 15.0(2)SE4 C2960-LANBASEK9-M
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE
(fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen
Press RETURN to get started!
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW5
SW5(config)#interface vlan 1
SW5(config-if)#ip address 192.168.2.8 255.255.255.0
SW5(config-if)#no shutdown
SW5(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
exit
SW5(config)#ip default-gateway 192.168.2.254
SW5(config)#end
SW5#
%SYS-5-CONFIG_I: Configured from console by console
Copy
Paste


Switch3
Physical
Config
CLI
Attributes
IOS Command Line Interface
Power supply serial number : AZS1007032H
Model revision number : B0
Motherboard revision number : B0
Model number : WS-C2960-24TT-L
System serial number : FOC1010X104
Top Assembly Part Number : 800-27221-02
Top Assembly Revision Number : A0
Version ID : V02
CLEI Code Number : COM3L00BRA
Hardware Board Revision Number : 0x01
Switch Ports Model SW Version SW Image
* 1 26 WS-C2960-24TT-L 15.0(2)SE4 C2960-LANBASEK9-M
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE
(fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen
Press RETURN to get started!
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW3
SW3(config)#interface vlan 1
SW3(config-if)#ip address 192.168.2.9 255.255.255.0
SW3(config-if)#no shutdown
SW3(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
exit
SW3(config)#ip default-gateway 192.168.2.254
SW3(config)#end
SW3#
%SYS-5-CONFIG_I: Configured from console by console
Copy
Paste

Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```
Power supply serial number       : AZS1007032H
Model revision number            : B0
Motherboard revision number      : B0
Model number                     : WS-C2960-24TT-L
System serial number             : FOC1010X104
Top Assembly Part Number         : 800-27221-02
Top Assembly Revision Number     : A0
Version ID                       : V02
CLEI Code Number                 : COM3L00BRA
Hardware Board Revision Number   : 0x01

Switch Ports Model               SW Version            SW Image
------ ----- -----               ----------            ----------
*    1 26    WS-C2960-24TT-L     15.0(2)SE4            C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE
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Press RETURN to get started!



Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW2
SW2(config)#interface vlan 1
SW2(config-if)#ip address 192.168.2.7 255.255.255.0
SW2(config-if)#no shutdown

SW2(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
exit
SW2(config)#ip default-gateway 192.168.2.254
SW2(config)#end
SW2#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
Power supply serial number       : AZS1007032H
Model revision number            : B0
Motherboard revision number      : B0
Model number                     : WS-C2960-24TT-L
System serial number             : FOC1010X104
Top Assembly Part Number         : 800-27221-02
Top Assembly Revision Number     : A0
Version ID                       : V02
CLEI Code Number                 : COM3L00BRA
Hardware Board Revision Number   : 0x01

Switch Ports Model               SW Version            SW Image
------ ----- -----               ----------            ----------
*    1 26    WS-C2960-24TT-L     15.0(2)SE4            C2960-LANBASEK9-M

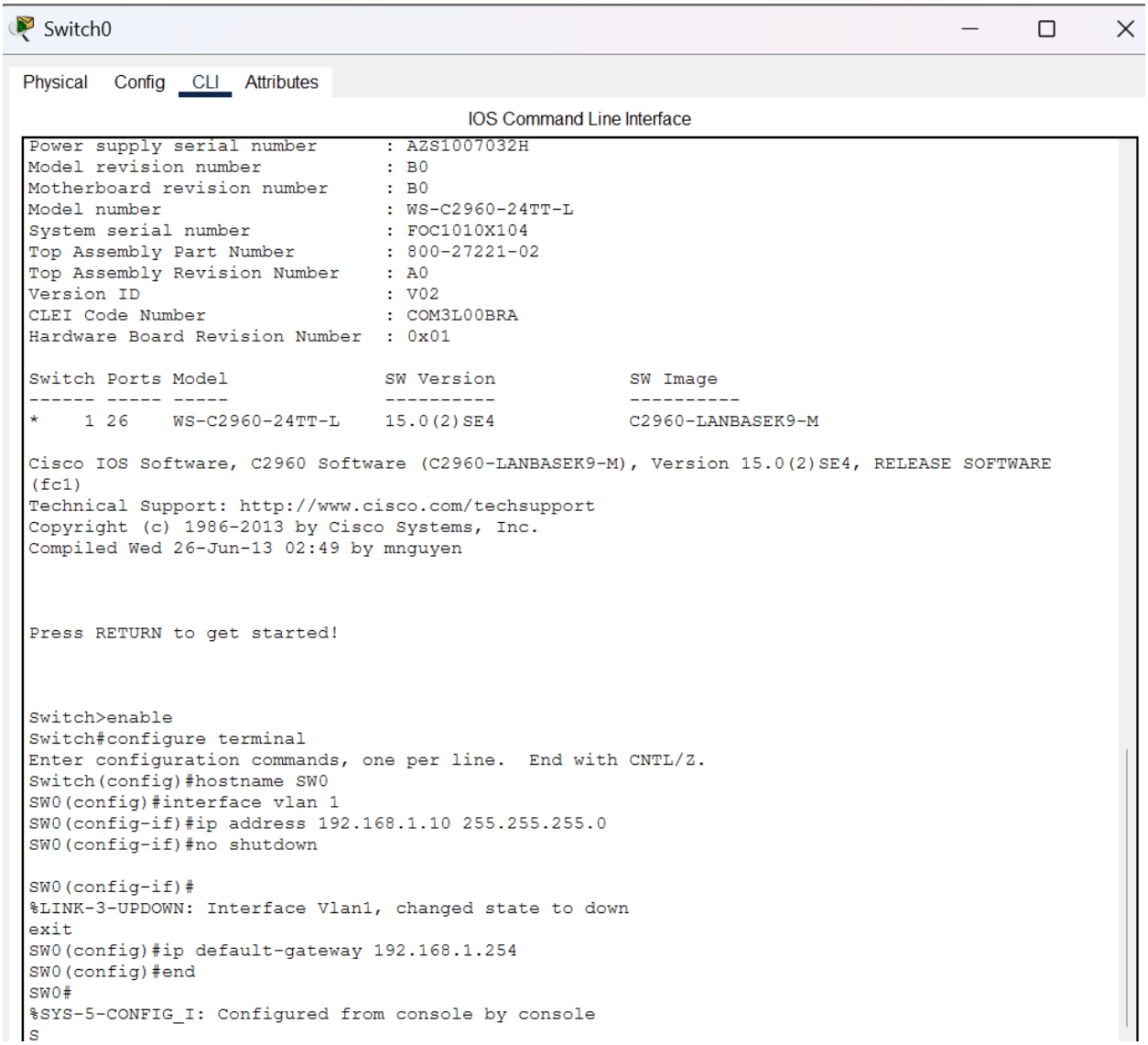
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE
(fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
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Press RETURN to get started!

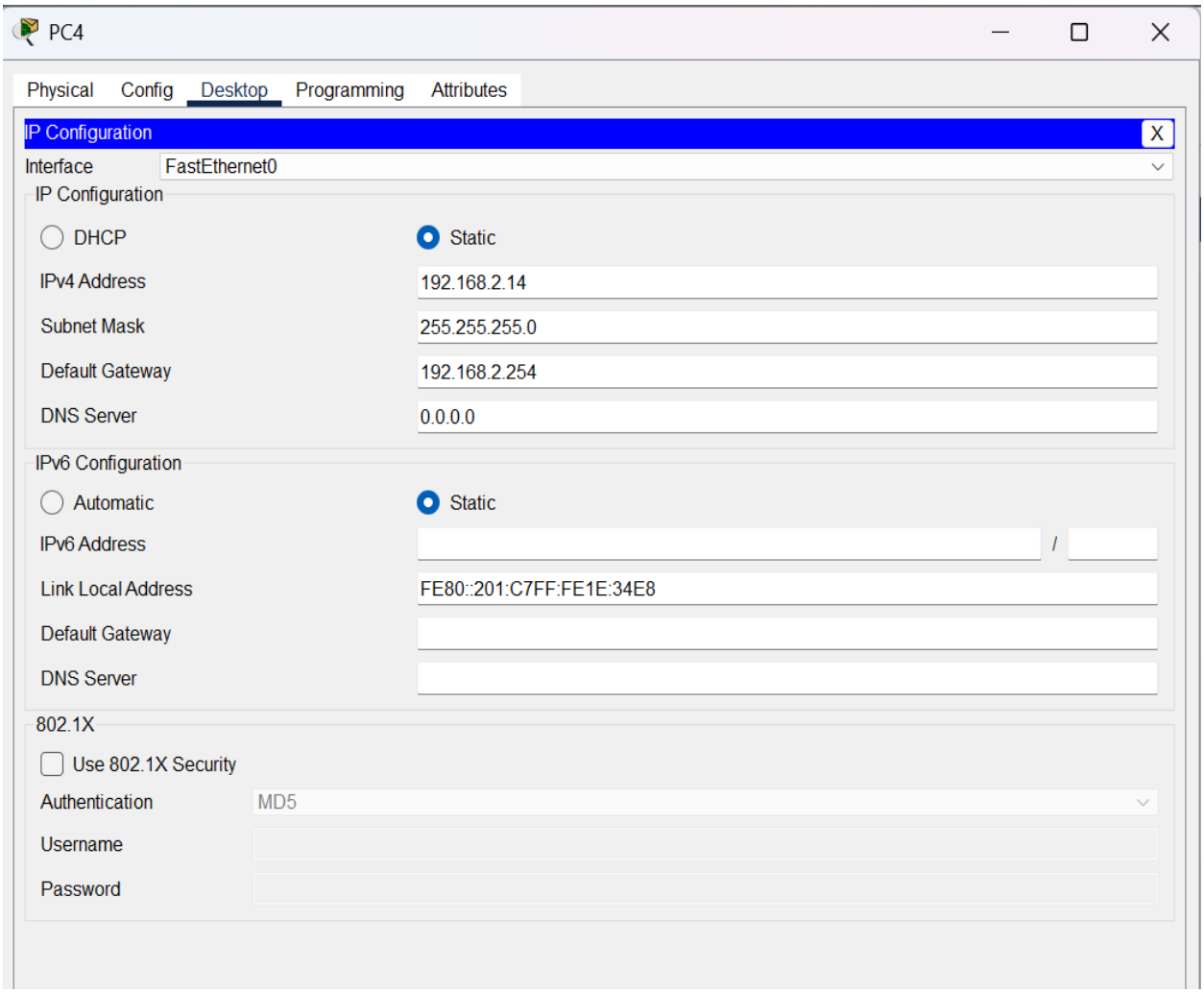


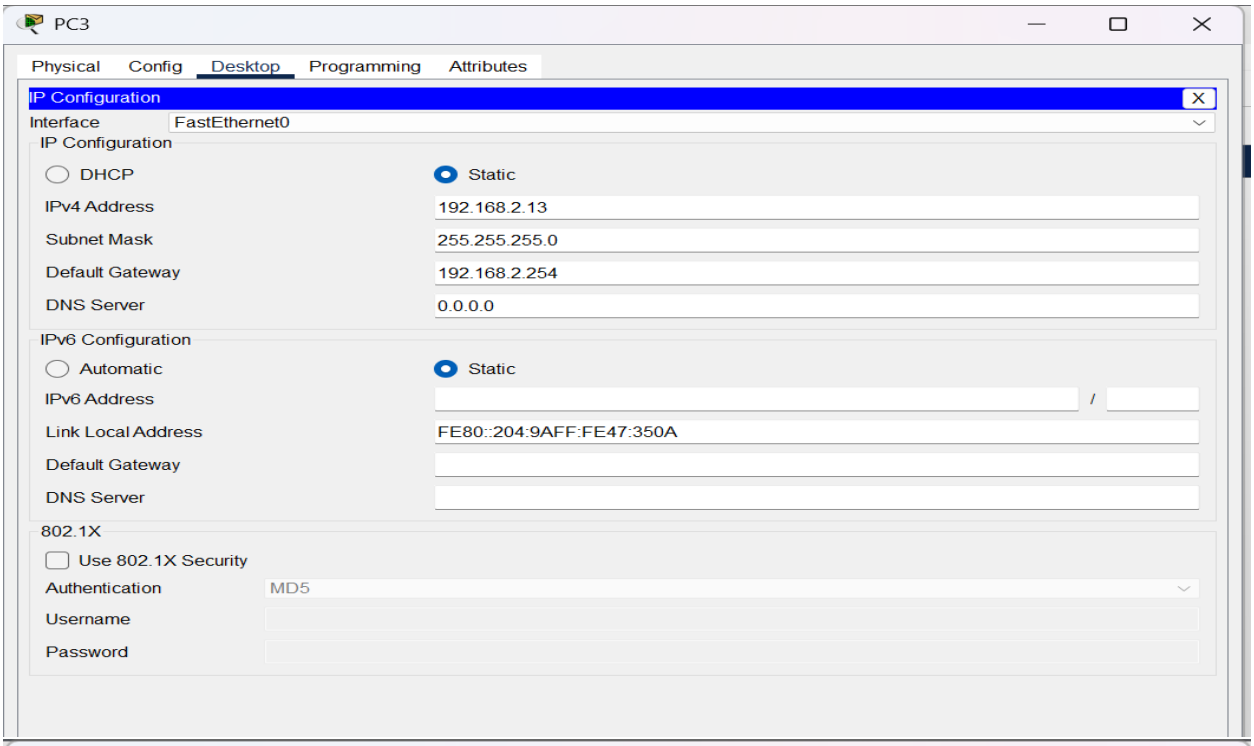
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW1
SW1(config)#interface vlan 1
SW1(config-if)#ip address 192.168.2.6 255.255.255.0
SW1(config-if)#no shutdown

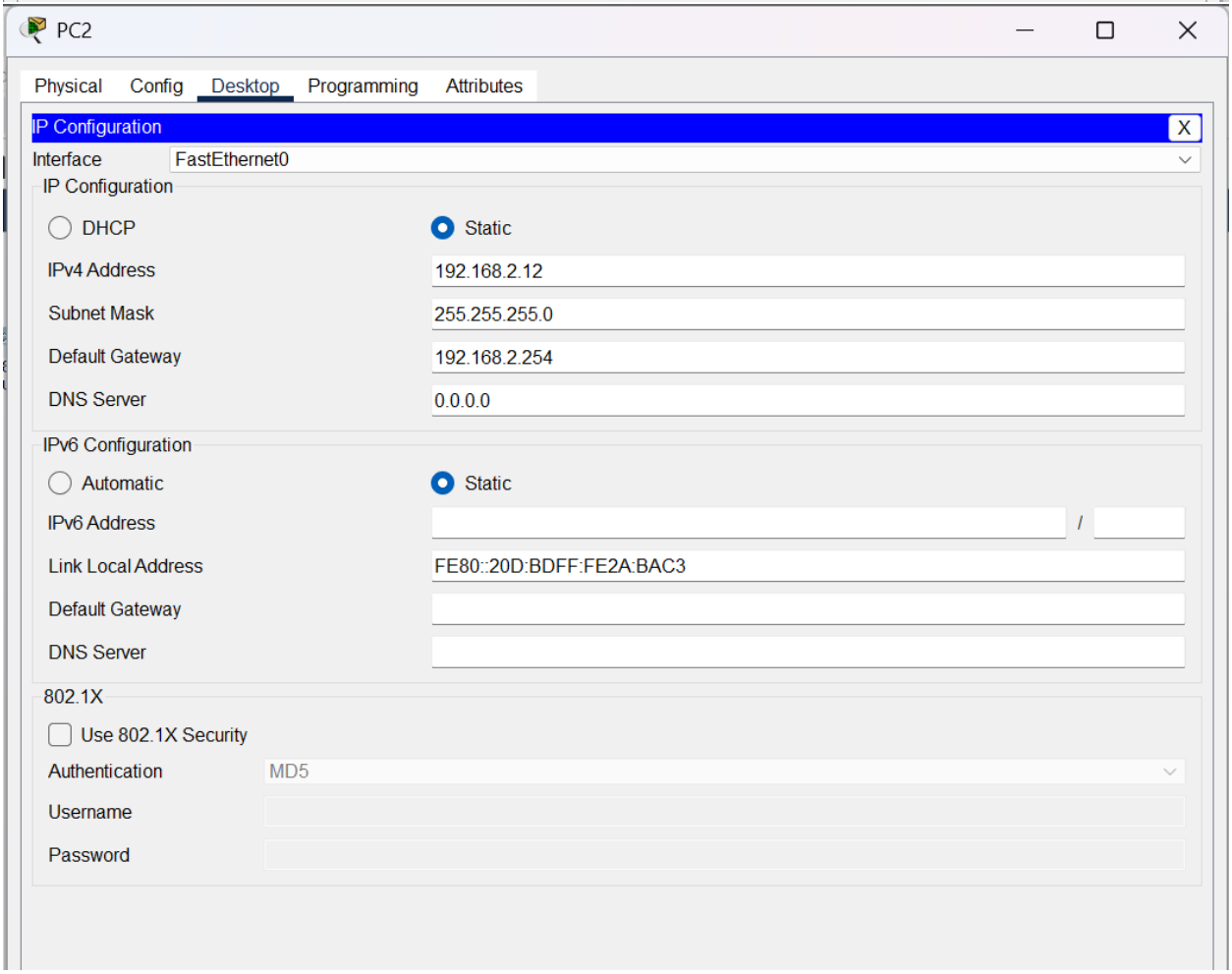
SW1(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
exit
SW1(config)#ip default-gateway 192.168.2.254
SW1(config)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console
```

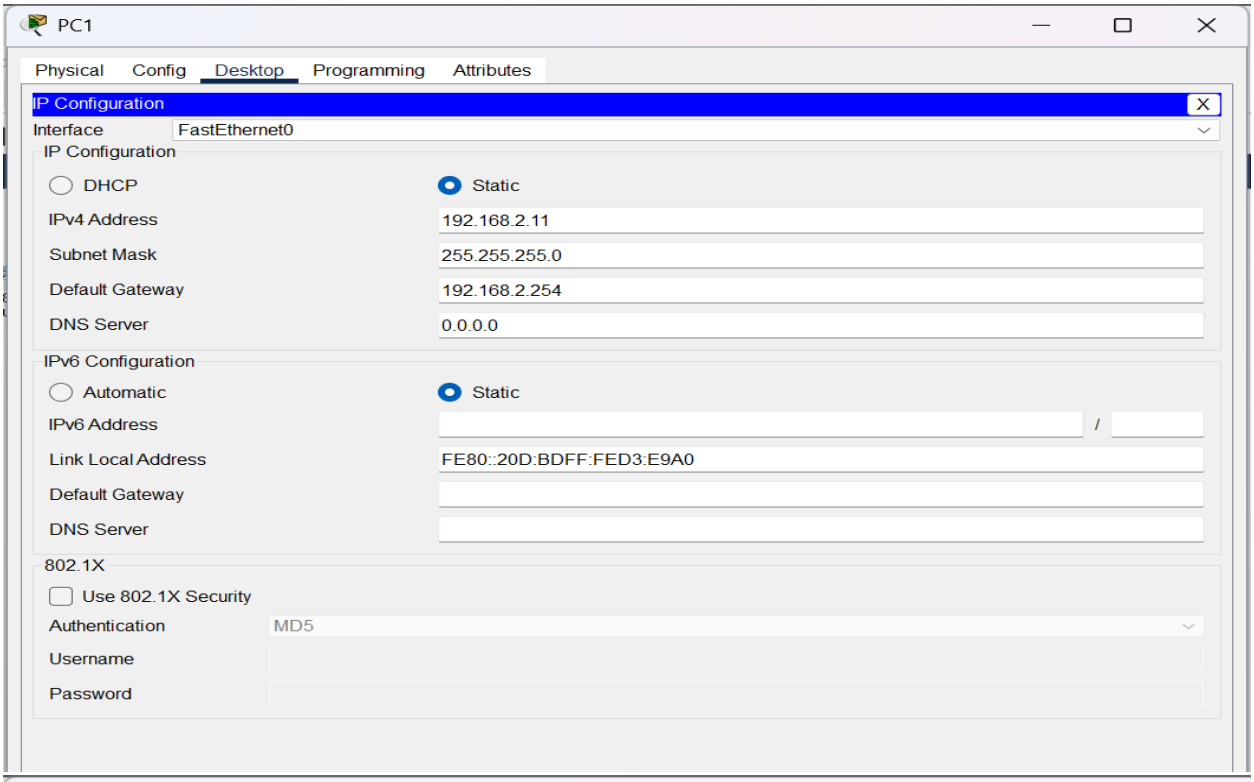

Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Power supply serial number : AZS1007032H
Model revision number : B0
Motherboard revision number : B0
Model number : WS-C2960-24TT-L
System serial number : FOC1010X104
Top Assembly Part Number : 800-27221-02
Top Assembly Revision Number : A0
Version ID : V02
CLEI Code Number : COM3L00BRA
Hardware Board Revision Number : 0x01
Switch Ports Model SW Version SW Image
* 1 26 WS-C2960-24TT-L 15.0(2)SE4 C2960-LANBASEK9-M
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(fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen
Press RETURN to get started!
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW0
SW0(config)#interface vlan 1
SW0(config-if)#ip address 192.168.1.10 255.255.255.0
SW0(config-if)#no shutdown
SW0(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
exit
SW0(config)#ip default-gateway 192.168.1.254
SW0(config)#end
SW0#
%SYS-5-CONFIG_I: Configured from console by console
S

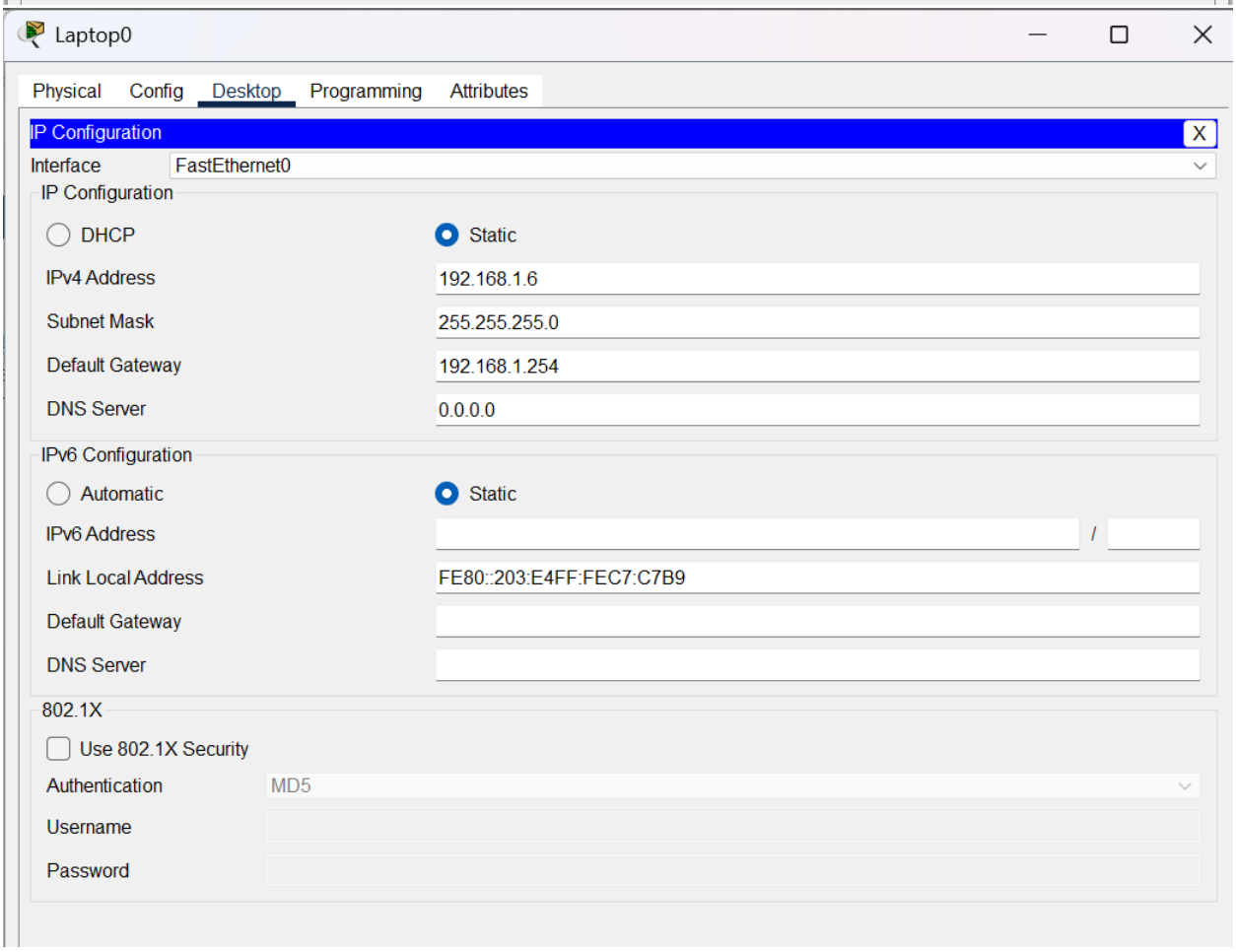
## Attribuons des adresses IPv4 aux périphériques

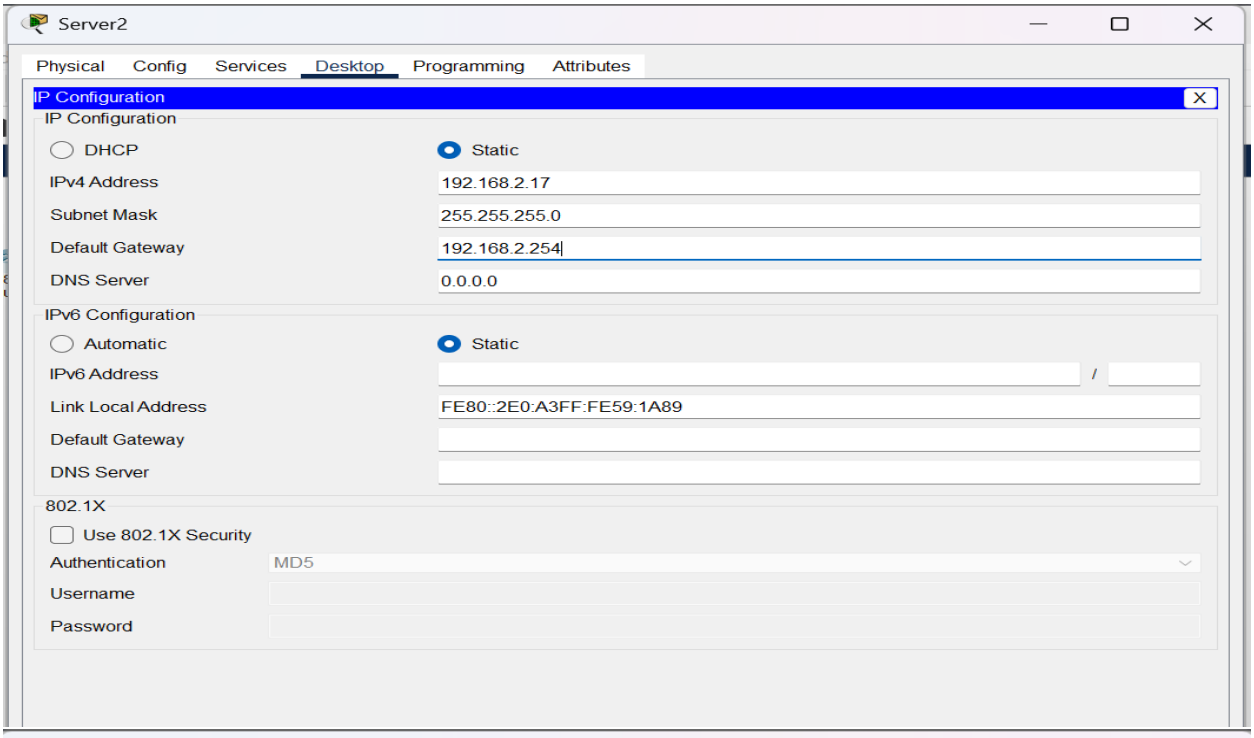


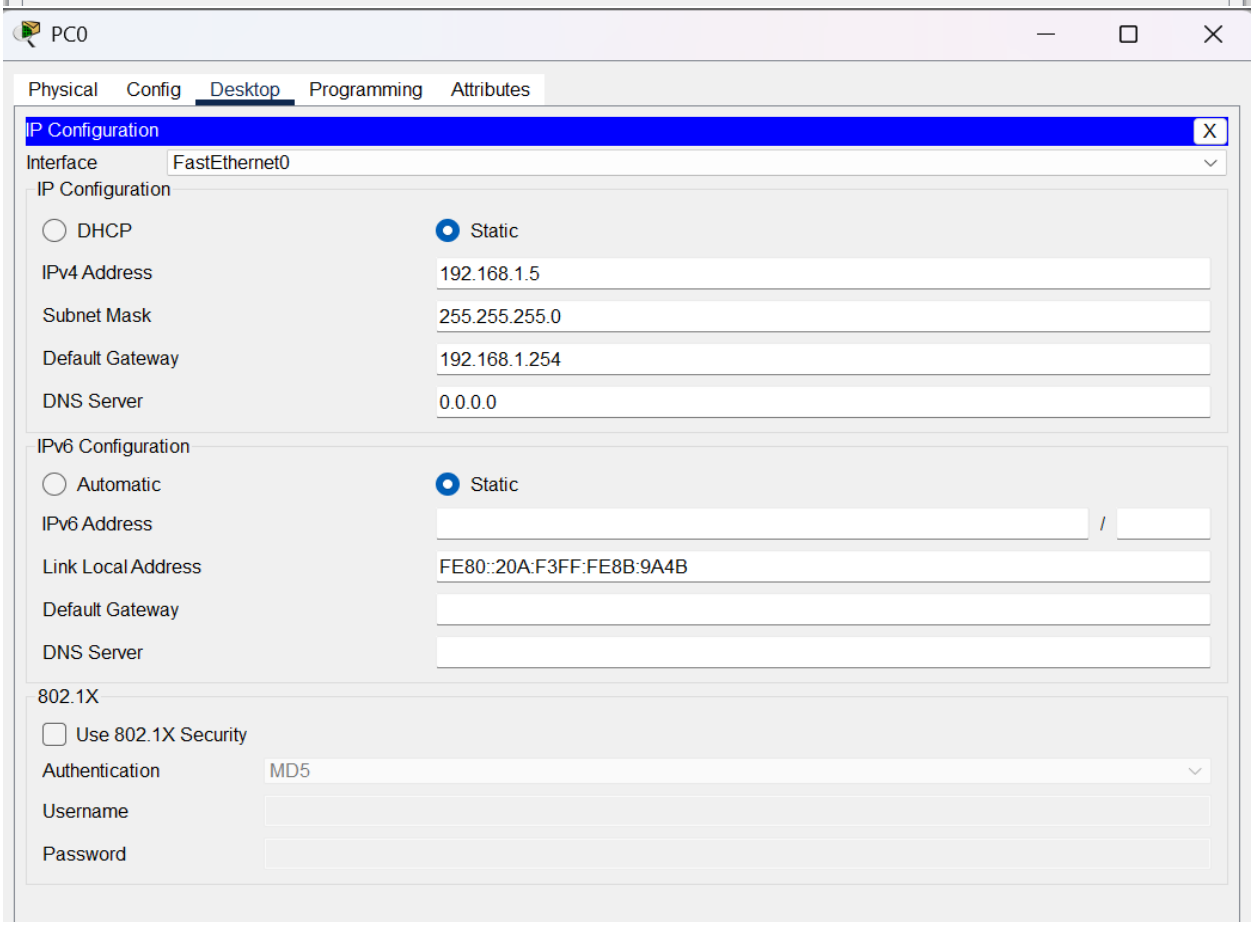

PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.13
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.254
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::204:9AFF:FE47:350A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

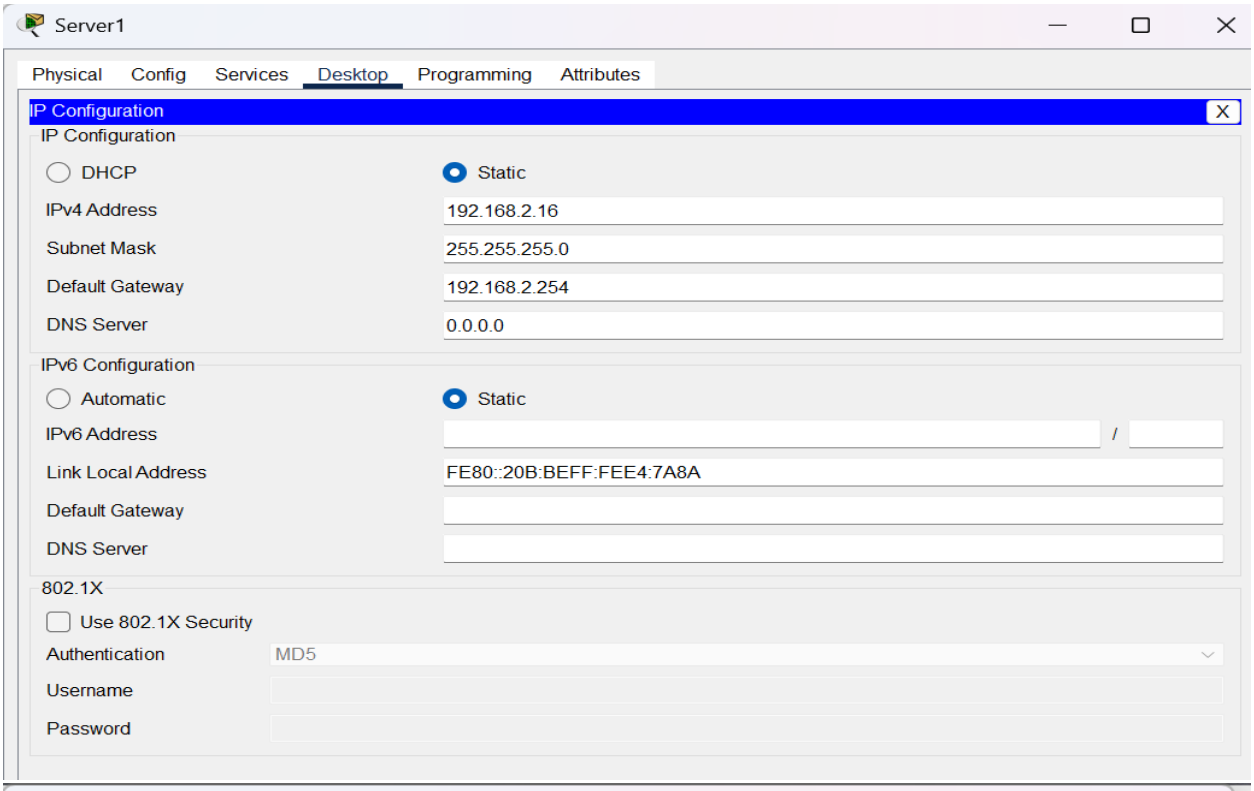

PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.12
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.254
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::20D:BDFF:FE2A:BAC3
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

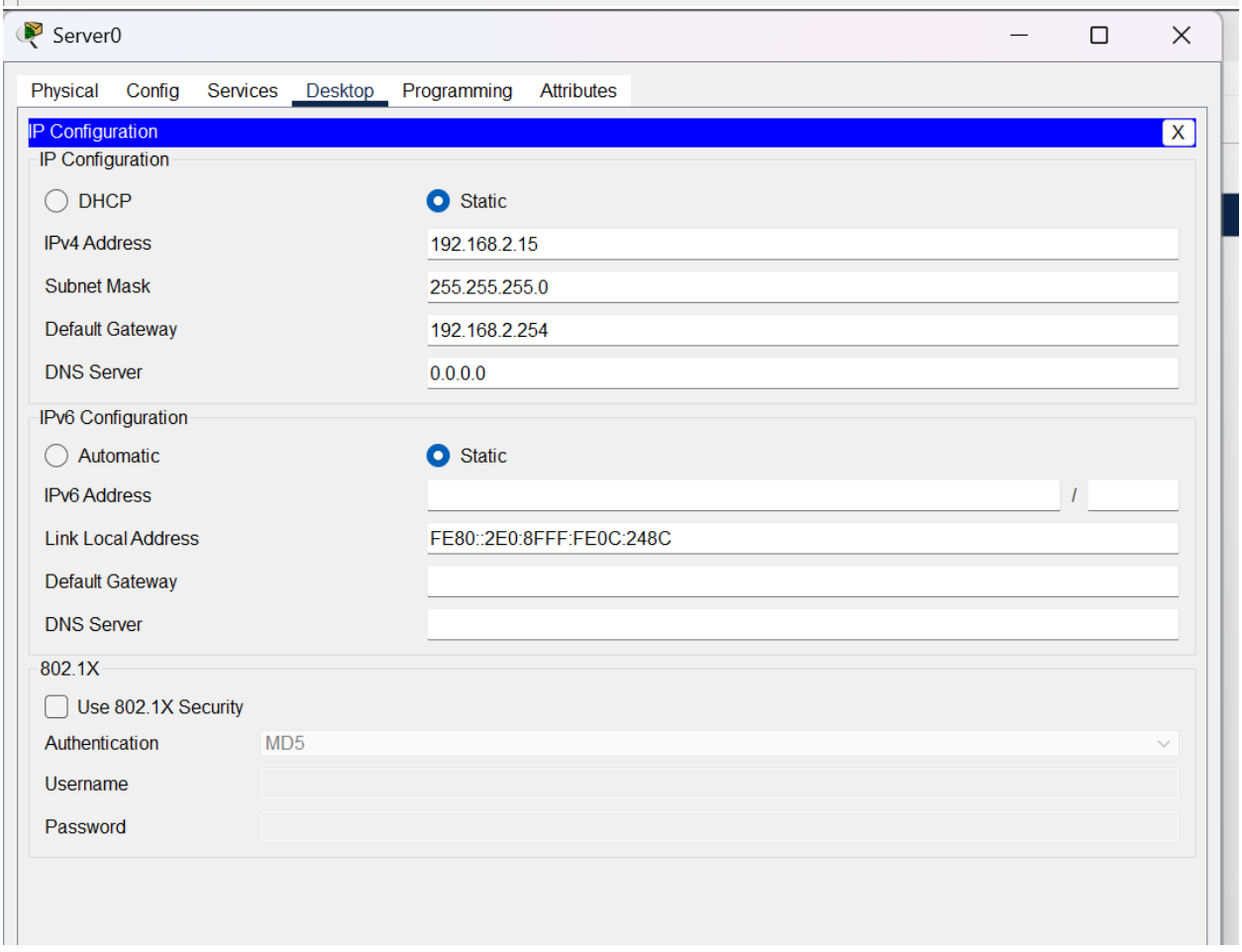

PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.11
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.254
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::20D:BDFF:FED3:E9A0
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

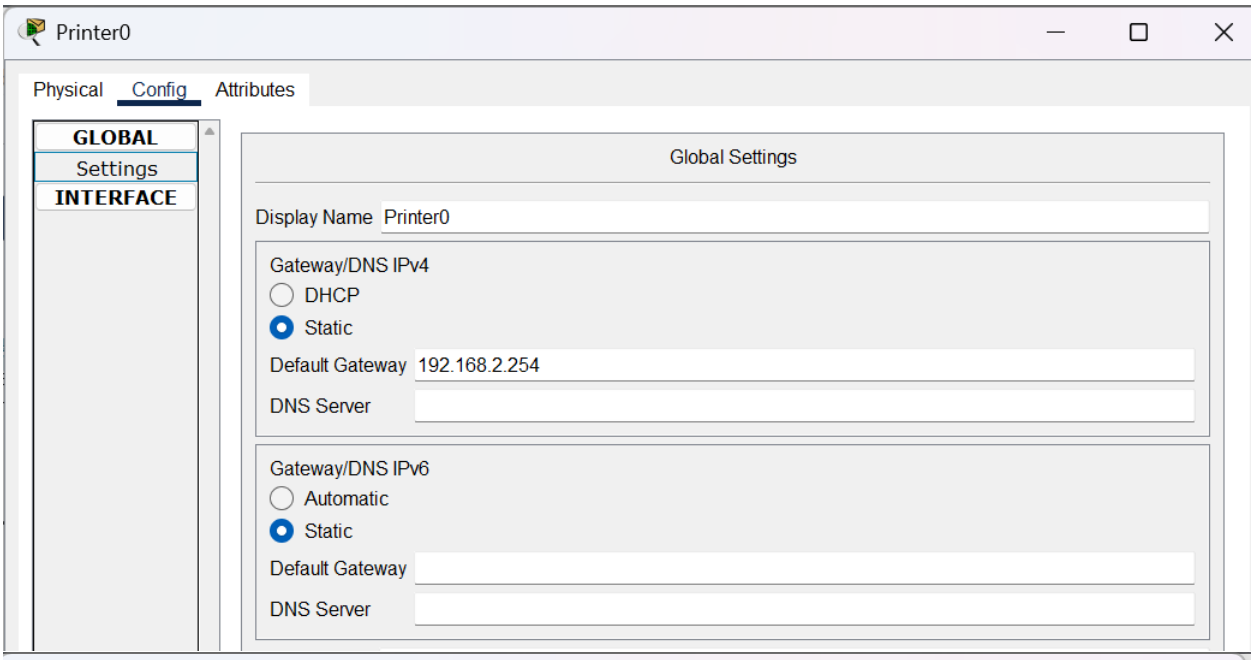

Laptop0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.6
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.254
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::203:E4FF:FEC7:C7B9
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

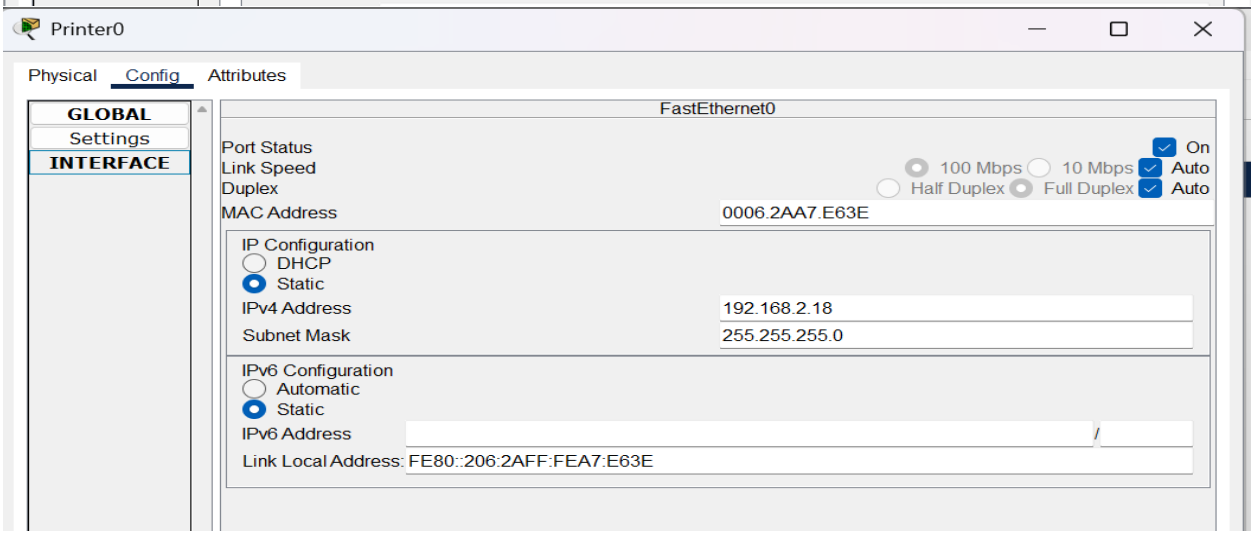

Server2
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.17
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.254
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2E0:A3FF:FE59:1A89
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

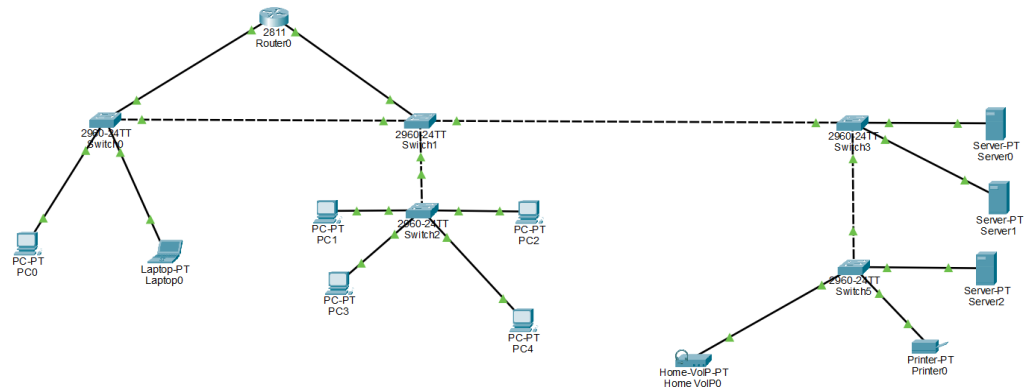

PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.5
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.254
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::20A:F3FF:FE8B:9A4B
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


Server1
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.16
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.254
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::20B:BEFF:FEE4:7A8A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

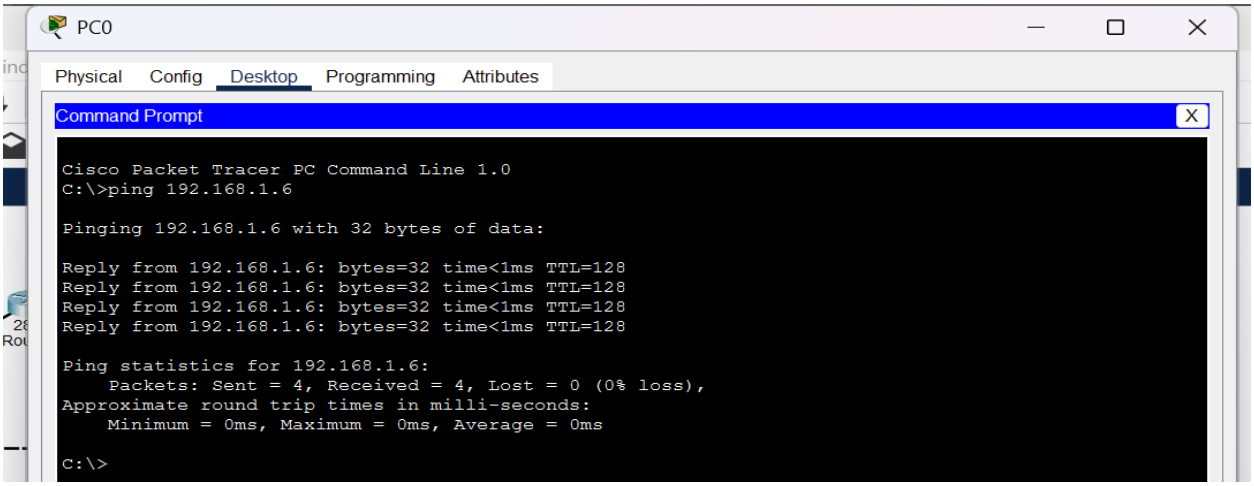

Server0
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.15
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.254
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2E0:8FFF:FE0C:248C
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

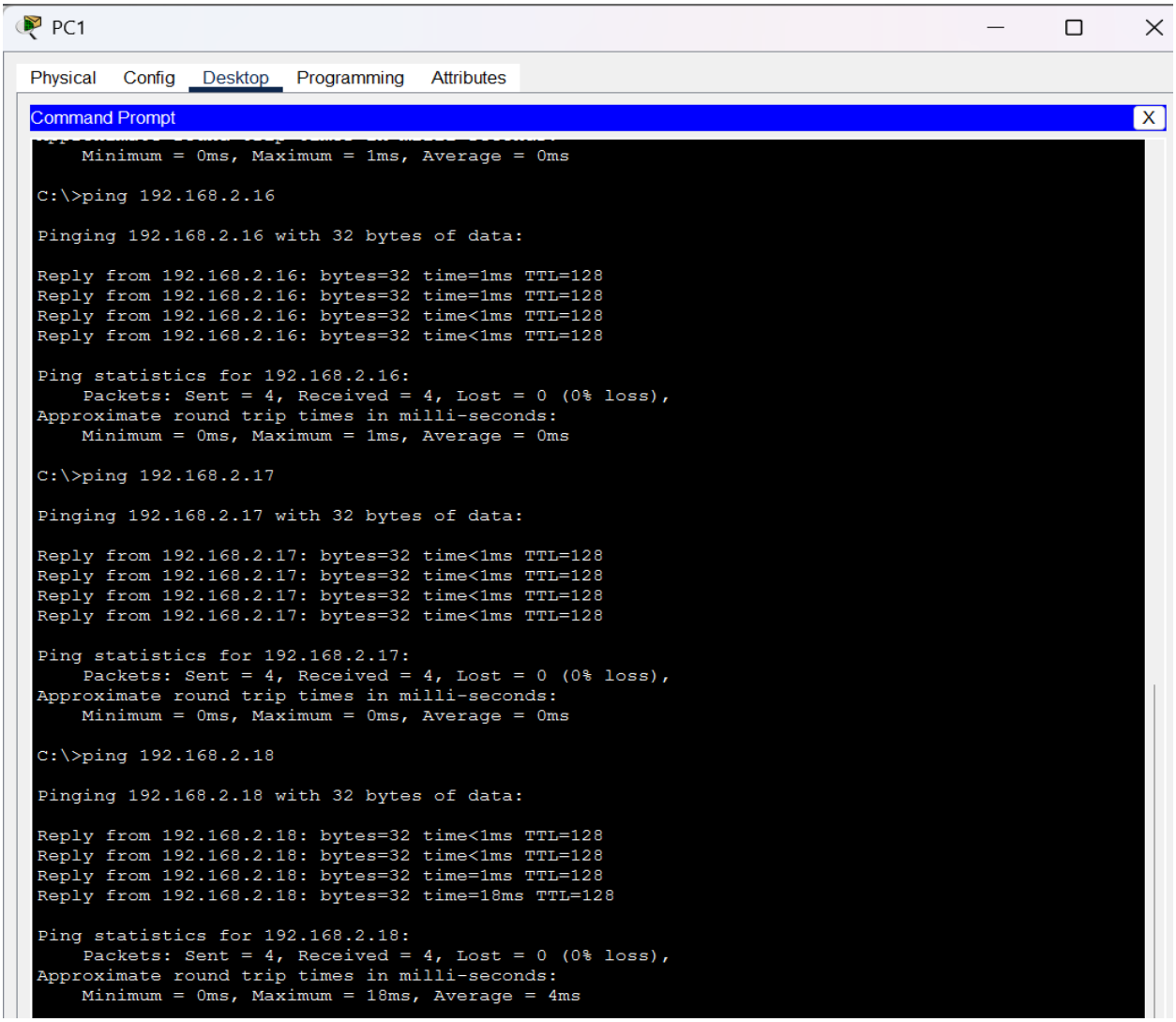

Printer0
Physical
Config
Attributes
GLOBAL
Settings
INTERFACE
Global Settings
Display Name
Printer0
Gateway/DNS IPv4
DHCP
Static
Default Gateway
192.168.2.254
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server


Printer0
Physical
Config
Attributes
GLOBAL
Settings
INTERFACE
FastEthernet0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0006.2AA7.E63E
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.18
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::206:2AFF:FEA7:E63E

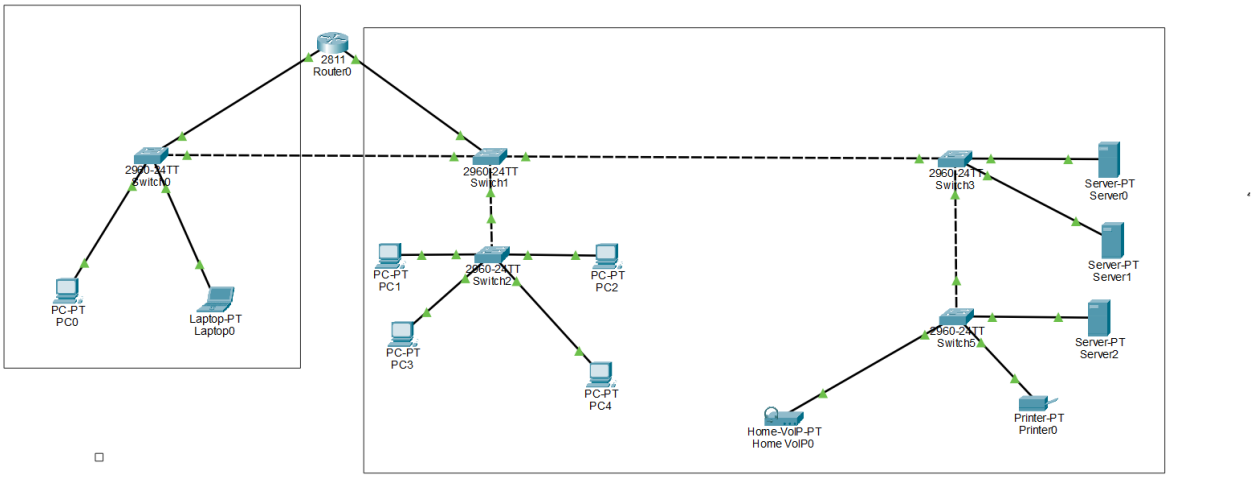

2811
Router0
2960-24TT
Switch0
2960-24TT
Switch1
2960-24TT
Switch3
Server-PT
Server0
Server-PT
Server1
PC-PT
PC0
Laptop-PT
Laptop0
PC-PT
PC1
2960-24TT
Switch2
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
2960-24TT
Switch4
Server-PT
Server2
Home-VoIP-PT
Home VoIP0
Printer-PT
Printer0

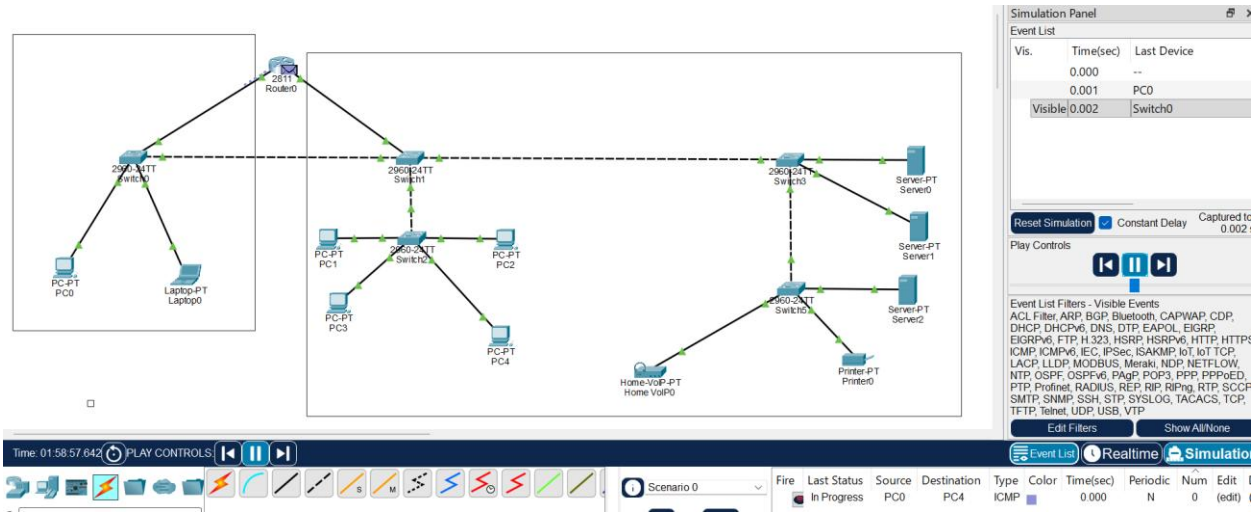
## Tester la connectivité entre les hôtes.

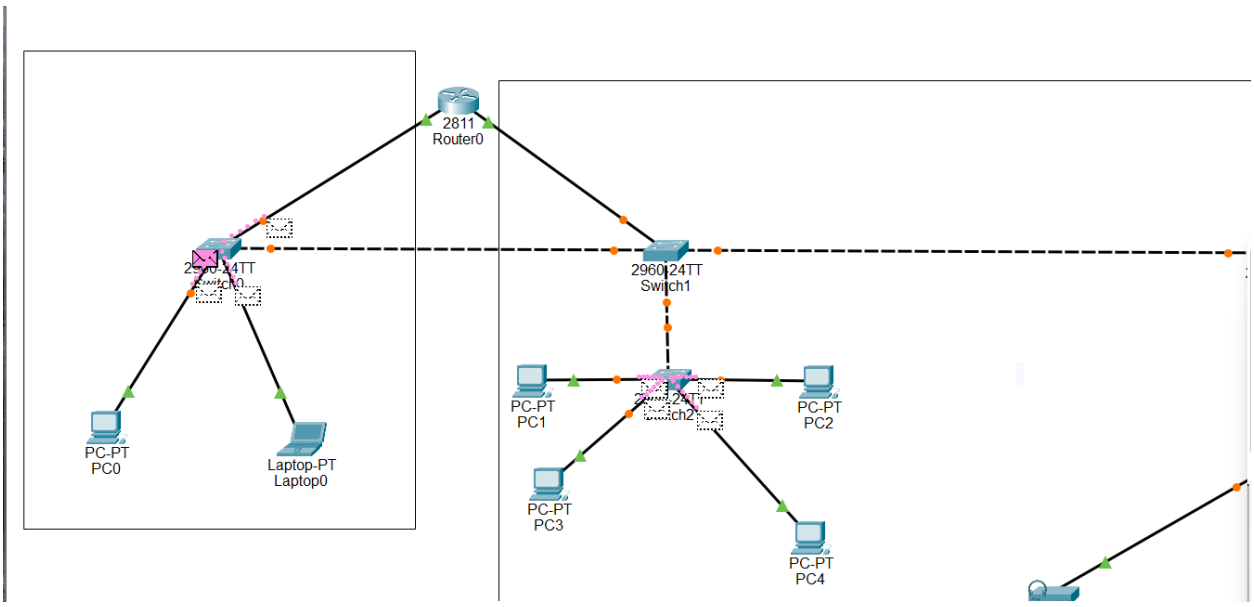


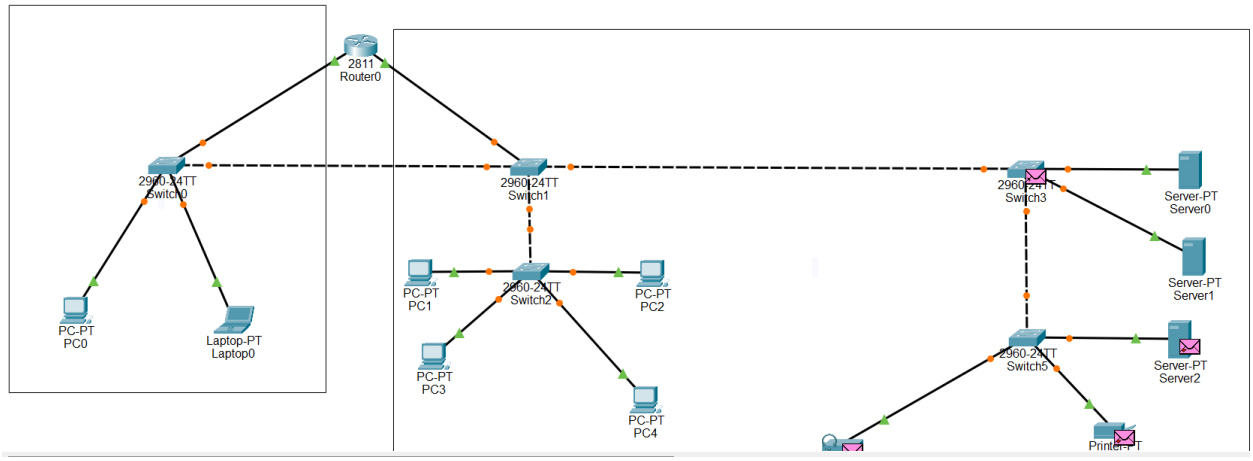



## Utiliser le Mode Simulation pour analyser le trafic réseau.






## 4. CONCLUSIONS

- **Acquis techniques :** Ce TD a permis de maîtriser la configuration initiale d'un commutateur Cisco 2960 et d'un routeur Cisco 2811 en mode CLI, l'adressage IPv4 ainsi que l'analyse dynamique des protocoles ARP, ICMP et NDP grâce au Mode Simulation de Packet Tracer.
- **Synthèse physique et logique :** L'observation des échanges en simulation illustre concrètement les notions du Module 4 : les données numériques encapsulées sont transmises sous forme de signaux électriques sur les câbles en cuivre avec une latence négligeable (< 1 ms) et sans perte de paquets lorsque les paramètres de liaison sont correctement configurés.
- **Difficultés rencontrées et solutions :** Lors de la configuration du routeur, deux contraintes ont été identifiées : l'état désactivé par défaut des interfaces (résolu par la commande `no shutdown`) et la nécessité d'attribuer des adresses IP valides appartenant à des sous-réseaux distincts sur `FastEthernet0/0` et `FastEthernet0/1` afin d'éviter tout chevauchement d'adressage.
- **Bilan :** L'ensemble des topologies et des tests de connectivité entre les hôtes et les deux zones réseau (Jacmel 1 et Jacmel 2) a été validé avec succès.